

SECTION 2 – ROOM LAYOUTS

TABLE OF CONTENTS

<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
2-1	INTRODUCTION	2-3
2-2	ROOM SIZES	2-3
2-3	MULTIPLE MR SYSTEMS SITE	2-6
2-3-1	Two Magnet Site Layout	2-6
2-3-2	Equipment Room Shared By Multiple MR Systems	2-7
2-4	MINIMUM DOOR/HALLWAY SIZES	2-7
2-5	CABLING CONSIDERATIONS	2-9
2-5-1	Floor Duct	2-9
2-5-2	Raceway	2-9
2-5-3	Raised Flooring	2-9
2-5-4	Conduit	2-10
2-6	FLOORING	2-10
2-6-1	Cable Routing Areas	2-10
2-6-2	Finished Floor	2-10
2-7	SPECIAL SITING CONSIDERATIONS	2-11
2-7-1	System Cooling Cabinet Subsystem Equipment	2-11
2-7-2	Pneumatic Patient Alert (PA1)	2-18
2-7-3	SPT Phantom Set Shipping/Storage Cabinet	2-18
2-7-4	Magnet Monitor Equipment	2-19
2-7-5	System Monitoring & Support Connectivity Requirements	2-20
2-7-6	Step-Up Transformers for 200/208 V Sites	2-20
2-8	ARCHITECTURAL REMINDERS	2-21
2-9	FLOOR LOADING AND WEIGHTS	2-23
2-9-1	Magnet Loading Considerations	2-25
2-9-2	Anchoring And Seismic Considerations	2-25
2-9-3	Operator Workspace Mounting Requirements	2-25
2-10	COMPONENT DIMENSIONS	2-26

2-1 INTRODUCTION

When laying out a floor plan there are special considerations that must be taken into account due to the magnetic field effect on certain medical implants (including cardiac pacemakers, neurostimulators, and biostimulation devices) and the environmental effect (motors, steel, etc.) on the field homogeneity. The maximum magnetic field in which the equipment can be located is listed in Table 2-1. Selected magnetic shielding of some devices and equipment is possible but must be handled on an individual basis. Refer to Section 6, INTERCONNECTION DATA, for cable length considerations.

The RF shielded room (Magnet Room) is unique in that the room must be shielded from outside radio frequency interference. This is done by enclosing the room with metal walls, floors and ceiling. These shielding requirements impose special considerations which are addressed in Section 7, RF SHIELDED ROOM.

The Magnet Room can be magnetically shielded to reduce the magnet fringe field or to shield the magnet from the effects of the external environment. Refer to Section 3-4, MAGNETIC SHIELDING, for magnet shielding considerations.

Note

The contents of this document do not support site plans for interventional procedures.

2-2 ROOM SIZES

Table 2-2 contains minimum room dimensions necessary for an MR suite. Table 2-2 also contains issues which are created by reduction in service access, operator access, and equipment space.

TABLE 2-1
PROXIMITY LIMITS

GAUSS (mT) LIMIT See Notes 1 & 2	EQUIPMENT	
0.5 GAUSS (0.05mT) OR LESS See Note 4	• Nuclear cameras	
1 GAUSS (0.1mT) OR LESS See Note 4	• Positron Emission Tomography scanner • Linear Accelerator • Cyclotrons • Accurate Measuring scale • Image intensifiers • Color TV	• Video display (color, B/W, monochrome) • CT scanner • Ultrasound • Lithotripter • Electron microscope • Advantage Workstation with CRT Monitor
3 GAUSS (0.3mT) OR LESS See Note 4	• Power transformers • Main electrical distribution transformers • 150 KVA 208 to 480Y277 Volt, 60 Hz transformer (option) • 150 KVA 200 to 400Y230 Volt, 50/60 Hz transformer (option)	• Moving steel equipment such as: – Fork lift trucks – Dumb waiters – Electric transport carts – Loading dock (truck traffic) – Elevators – Escalators – Helicopters (See Note 3)
5 GAUSS (0.5mT) OR LESS	• Cardiac pacemakers • Neurostimulators	• Biostimulation devices
10 GAUSS (1mT) OR LESS	• Magnetic tapes and floppy discs • Hard copy imagers • Line printers • Video Cassette Recorder (VCR) • Film processor • Credit cards, watches, and clocks • Telephone switching station • Water cooling equipment	• HVAC equipment • Major mechanical equipment room • Large steel equipment such as: – Emergency generators – Commercial laundry equipment – Food preparation area – Air conditioning chiller – Fuel storage tanks – Motors greater than 5 horsepower • X-ray tubes
30 GAUSS (3mT) OR LESS	• System Cabinet (MR2) • System Cooling Cabinet (SCC)	• System Cooling Auxiliary Cabinet (SCC2) • Remote Condenser Unit (RCU) for SCC
50 GAUSS (5mT) OR LESS	• Operator Workspace Cabinet (OW1A2) • LCD Color Monitor (OW1A6) (See note 5) • Main Disconnect Panel (MDP) • Magnet Monitor (MSM1)	• UPS for Magnet Monitor (MSM4) • Patient Comfort Compressor (PCC) • Telephones • Metal detector for screening
100 GAUSS (10mT) OR LESS	• Cryo Cooler Compressor Cabinet (MS5) • Shield Cooler Compressor Cabinet (MS8)	• Magnet Service Tool Power Supply Cabinet • Pneumatic Patient Alert Control Box (PA1)
200 GAUSS (20mT) OR LESS	• Penetration Panel (PP1) • Power Cabinet (MR1)	• Magnet Rundown Unit (MS4)

- Note**
- 1 Refer to SECTION 3, MAGNETIC FIELD CONSIDERATIONS, for magnet field plots.
 - 2 Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable.
 - 3 Verify operating limits with provider of helicopter service
 - 4 Refer to **Section 4-13 CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS** for additional Active Shield magnet proximity limits.
 - 5 If gauss limit is more than indicated for OW with LCD Color Monitor then contact GE to determine impacts on OW equipment.

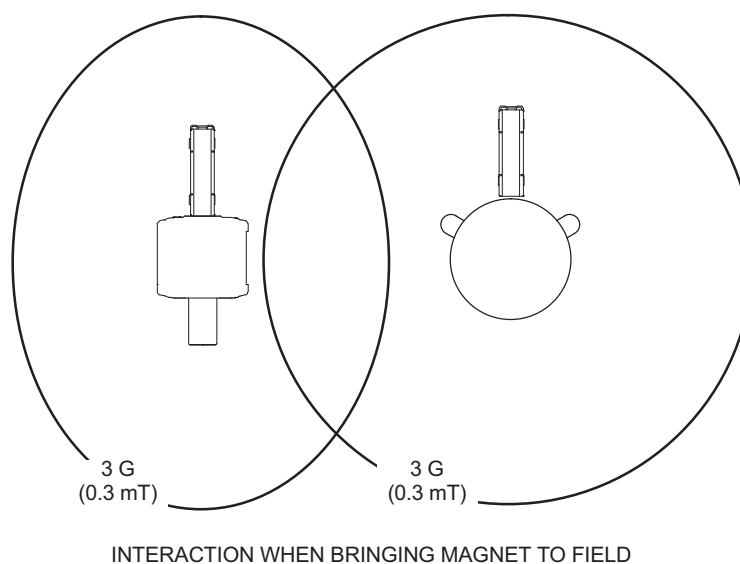
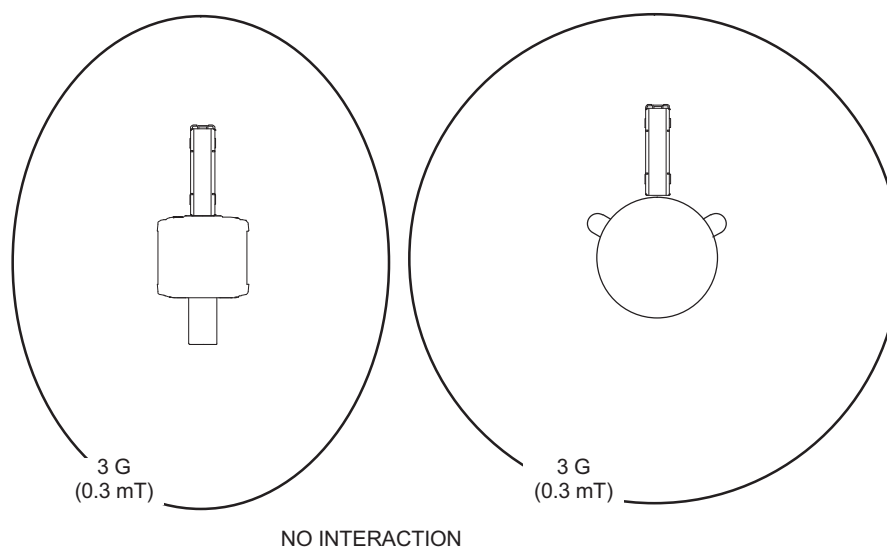
TABLE 2-2
MINIMUM ROOM INSIDE CLEAR SPACE DIMENSIONS
 (VALUES DO NOT INCLUDE MAGNETIC AND/OR RF SHIELDING OR FINISHED WALL)

ROOM	0.7T SYSTEM CONFIGURATION MINIMUM VALUES	
MAGNET ROOM See Note 1 Width X Depth: ft–in. (m) Area: ft ² (m ²) Ceiling Height: ft–in. (m) See Note 4	13.12 x 19.68 (4 x 6) See Note 3 258.20 (24.00) 8–7 (2.62)	
EQUIPMENT ROOM See Note 2 Width X Depth: ft–in. (m) Area: ft ² (m ²)	14 x 9 (4.27 x 2.74) 126 (11.70)	
CONTROL ROOM Width X Depth: ft–in. (m) Area: ft ² (m ²)	5* x 7 (1.52* x 2.13) 35 (3.24)	
TOTAL SYSTEM AREA: ft ² (m ²)	419.2 (38.94)	
Note 1 Absolute Minimum Magnet Room dimensions will result in limited operator clearances and increased Magnet Service time. 2 Absolute Minimum Equipment Room and Control Room dimensions do not permit placement of air conditioning in room. Nor do they permit space for any optional equipment such as Laser Camera, etc. 3 Magnet Room dimensions do not contain 5 gauss line to room. 4 For Slab On Grade Sites the 0.7T magnet is limited to minimum of 18 inch (457 mm) thick concrete slab on grade installation as required in Section 7–6–4 Floors . For Suspension Foundation Sites the magnet requires a 2 inch (51 mm) thick Magnet Mounting Plate be recessed into the floor, refer to Section 7–7 MAGNET MOUNTING REQUIREMENTS INSIDE RF SHIELDED ROOM for additional floor requirements * Width is dependent on Magnet Room door location and customer’s approval of limited space available for operator.		

2-3 MULTIPLE MR SYSTEMS SITE

2-3-1 TWO MAGNET SITE LAYOUT

For two magnet installations interaction can occur between the magnetic fields. For 2 magnets not to interact at all (including when bringing magnet to field) the 3 gauss lines of each magnet must not intersect. If the 3 gauss lines intersect but remain outside each magnet's cryostat there will be interaction between the magnets when bringing to field. The orientation of the magnets is irrelevant. Consult the MR Siting & Shielding group for closer proximity of magnets.



TWO MAGNET INSTALLATION

ILLUSTRATION 2-1

2-3-2 Equipment Room Shared By Multiple MR Systems

When the Equipment Room is shared by two or more MR systems of the same field strength there is a potential for cross-talk of RF energy between the MR systems. RF cross-talk may cause noise artifacts in images. Proper planning and installation of the multiple systems in the shared Equipment Room can reduce the potential for cross-talk.

- The RF Screen of the Magnet Room for each system needs to meet the RF Attenuation specifications in **Section 7-1 RF SHIELDED ROOM SPECIFICATION**.
- The Equipment Room design, layout, and installation must meet the requirements of **APPENDIX C – REQUIREMENTS FOR EQUIPMENT ROOM SHARED BY MULTIPLE MR SYSTEMS** for:
 - Relative location of certain equipment cabinets of one MR system relative to other MR system(s) equipment cabinets.
 - Separation of the Penetration Panel of one MR system relative to the Penetration Panel of other MR system(s).
 - Separation of interconnects of one MR system relative to interconnects other system(s) and the managing of any excess RF Receive and RF Transmit cables for each MR System.

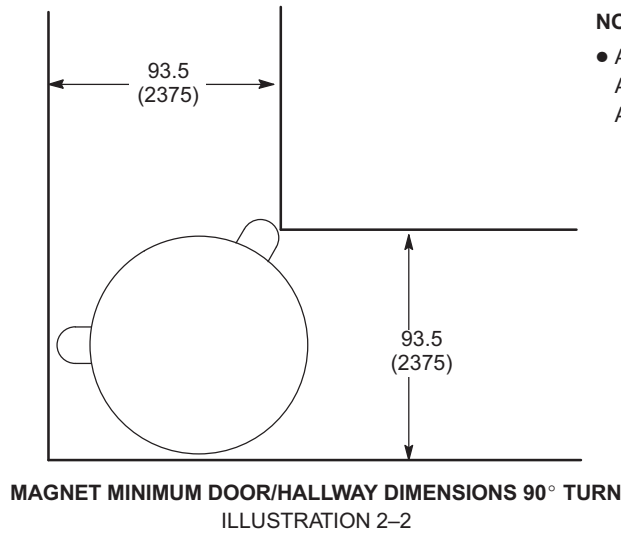
2-4 MINIMUM DOOR/HALLWAY SIZES

Table 2-3 lists minimum actual clearance opening dimensions for doors and hallways required by Signa equipment. Installation or replacement of components listed in Table 2-4 must be taken into consideration when determining hallway and door dimensions. Clearance for maneuvering around corners or turns must also be taken into consideration. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for Signa Component shipping dimensions.

TABLE 2-3
MINIMUM HALLWAY/DOOR DIMENSIONS

COMPONENT	MINIMUM HALLWAY/ DOOR WIDTH* in. (mm)	MINIMUM HALLWAY/ DOOR HEIGHT* in. (mm)	COMMENTS
Operator Workspace Table	32 (813)	80 (2032)	
System Cooling Cabinet	40 (1016) See Comments	76 (1930)	Minimum Hallway/Door width for 90° turn is 50 inches (1270 mm).
Equipment Cabinets	36 (914)	80 (2032)	
Cryogen delivery route and Storage Room	43 (1092)	80 (2032)	Width requirements due to size of 500 liter dewars. Width and height requirements vary dependent on the dewars used. Check with cryogen supplier.
Magnet	Refer to Note 1	Refer to Note 2	Refer to Section 8, SHIPPING AND DELIVERY DATA, for dimensions, illustrations and weights.
RF Room Door	Refer to Section 7, RF SHIELDED ROOM	Refer to Section 7, RF SHIELDED ROOM	
Note * Minimum hallway and door dimensions are actual clearance openings inside of the door casing. Width and height of rigging equipment is not included in above dimension.			
1 Minimum width depends on access route to removable panels of RF shielded room wall. For straight path (i.e. no bends or turns) recommended to allow 6 in. (153 mm) on both sides of magnet. Appropriate calculations must be performed if turns exist along proposed magnet delivery route. Illustration 2-2 shows dimensions for 90° turn.			
2 Final dimension is dependent on rigger equipment used, refer to SECTION 8, SHIPPING AND DELIVERY DATA.			

2-4 MINIMUM DOOR/HALLWAY SIZES (continued)



NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

TABLE 2-4
COMPONENT DIMENSIONS FOR INSTALLATION/REPLACEMENT

COMPONENT	APPROXIMATE WEIGHT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	COMMENTS
Magnet (uncrated)	Refer to comments.	Refer to comments.	Refer to Section 8, SHIPPING AND DELIVERY DATA, for dimensions, illustrations and weights.
System Cooling Cabinet (SCC)	1350 (614) dry	65 x 32.5 x 75.44 (1600 x 825.5 x 1916)	Cryo Cooler Compressor Cabinet & Shield Cooler Compressor Cabinet are installed in lower portion of System Cooling Cabinet at site.
System Cooling Auxiliary Cabinet (SCC2)	550 (250) dry	34.12 x 40.5 x 42.12 (867 x 1029 x 1070)	SCC2 must be located immediately next to SCC. SCC2 can be positioned on either side of SCC.
High Pressure Blower (SCC2)	150 (69) dry	32 x 40.5 x 20 (813 x 1016 x 508)	SCC2 will be installed on top of the SCC.
RF Coil	160 (73)	12 x 36 x 42 (305 x 914 x 1067)	The RF Coil (upper & lower) are shipped separate from the magnet and installed at site.
Replacement Gradient Coil in Shipping Box	350 (159)	42 x 42 x 12 (1067 x 1067 x 305)	Initial Gradient Coil Assembly is shipped installed in the Magnet. Shipping box is used for replacement coil assembly only.
Gradient Coil Replacement Tool Kit Crate	250 (114)	30 x 86 x 28 (762 x 2184 x 711)	The Gradient Coil Replacement Tool Kit is shipped in a wooden crate on casters.

2-5 CABLING CONSIDERATIONS

Several different methods for running cables are listed below and the customer should carefully consider the advantages and disadvantages of each.

Note

Customer current and future system utilization should be considered when determining method of running cables, i.e. surface floor duct or access floor behind magnet may not be acceptable if customer access is desired around the entire magnet.

Care must be taken to protect interconnecting cords and cables from physical damage (including water). Branch circuit conductors must be enclosed in metal raceway or metal wireway when concealed or when installed under raised flooring.

Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

Note

Only non-magnetic metal material (e.g. aluminum) can be used when routing cables in the Magnet Room.

2-5-1 Floor Duct

Recessed floor duct has advantages when used within a single room or two adjacent rooms. Floor duct combines a neat functional appearance with accessibility and room for expansion. The disadvantage is the amount of work required to install it, which is generally prohibitive in old installations. Floor ducts can be used in the Magnet Room, however, they must meet the requirements in **Section 7, RF SHIELDED ROOM**.

2-5-2 Raceway

Raceways offer some unique advantages when routing cables. It is very practical to use in existing structures since it is surface-mounted. There is no problem with pre-terminated cables since the entire raceway system can be opened. Raceway systems are relatively easy to expand as compared to other means of routing cables. However, surface-mounted raceways are not recommended for routing cables within the Magnet Room due to the number/size of cables and the trip hazard of the raceway.

2-5-3 Raised Flooring

Raised flooring is recommended for use in both the Equipment and Magnet Rooms due to the number and size of cables in the system. Cable accessibility and ease of alteration are just a few advantages of using raised flooring. Floor duct with dividers placed above the Magnet Room floor but beneath the raised flooring is a convenient method of separating electrical lines from water lines. However, if the area under the raised flooring is used for an air plenum, cables may have to be in raceway depending on local and national codes. **Note**, the Signa system interconnecting cables in the Equipment room are FT4 rated, not air plenum rated.

2-5-4 Conduit

Conduit has some important restrictions when used with a MR system. The primary problem is that the majority of cables used are pre-terminated, which greatly simplifies interconnection, but makes cabling difficult because of the added dimensions of the connectors. As a consequence, conduit size must allow for the dimension of the connectors and the possibility of additional cables being added as the system is upgraded in the future. Always size the conduit to allow the cable to pass through with all other cables already in the conduit. Conduit should not be used for running the main GE MR system cables in the Magnet Room due to the number and size of conduits needed.

Note

MR personnel must have an unobstructed path from the patient table to the area directly behind the Magnet. Therefore cable routing methods must not interfere with this pathway.

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes.

2-6 FLOORING

2-6-1 Cable Routing Areas

Use of a raised floor with covering to minimize static discharge is recommended in the Equipment Room and the portion of the Magnet Room for cable routing. For safety purposes in the Magnet Room, it is required that the raised flooring be made of aluminum. Depending on local and national codes, the area under the raised floor may possibly be used as an air conditioning plenum. If the area under the raised floor is to be used as an air plenum and for cable routing, 10 in. (254mm) of clear space from the underside of the raised floor to the permanent floor is recommended. Cabling, plumbing (water lines), etc. routed under the raised floor may affect air flow and needs to be considered if used as an air conditioning plenum. Also check local and national codes for fire protection requirements under raised floor.

Note

The Signa system interconnecting cables in the Equipment room are FT4 rated, not air plenum rated.

Ensure that the raised flooring, if used, can support the equipment and any transport device needed to move the equipment.

2-6-2 Finished Floor

The finished floor in the Magnet Room should be waterproof and be a conductive type flooring to reduce the possibility of a static discharge. Hard surface finished flooring is required in the Magnet Room for operation of the Swing Patient Table. Information on RF shielded room floor requirements can be found in Section 7, RF SHIELDED ROOM.

If carpeting is used in the Control Room, it should either be anti-static carpeting or treated with an anti-static solution. Carpeting is not recommended in cryogen storage area as well as along the dewar delivery route due to possible problems with moving the dewars.

2-7 SPECIAL SITING CONSIDERATIONS

The following system equipment have special siting concerns which need to be considered.

2-7-1 System Cooling Cabinet Subsystem Equipment

The System Cooling Cabinet (SCC) is a dedicated chiller system providing water cooling for

- Cryo Cooler Compressor Cabinet (MS5)
- Shield Cooler Compressor Cabinet (MS8)
- Gradient Coils
- System Cooling Auxiliary Cabinet (SCC2) which provides compressed cooled air to the RF Coil. The SCC and SCC2 **must** be located side-by-side.

The SCC is available in the 3 configuration options: 1) Air Cooled SCC with Remote Condenser Unit (RCU), 2) Water Cooled SCC, or 3) Air Cooled SCC. Each configuration is available configured for high voltage power (480Y/277 VAC, 400Y/230 VAC, 50/60 Hz). Low voltage (200 or 208 VAC) sites will need to provide a step-up transformer as described in **Section 2-7-6 Step-Up Transformers for 200/208 V Sites**. See Illustration 2-16 for System Cooling Cabinet (SCC), Illustration 2-18 for System Cooling Auxiliary Cabinet (SCC2), and Illustration 2-13 for Remote Condenser Unit (RCU).

Note

The system interconnects are designed for the Cryo Cooler Compressor Cabinet (MS5) and the Shield Cooler Compressor Cabinet (MS8) are to be installed in the lower portion of the System Cooling Cabinet (SCC) as shown in Illustrations 2-12, 2-15, and 2-16.



Continuous water cooling is critical for the Shield/Cryo Cooler Compressors and therefore MUST be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival. The System Cooling Cabinet (SCC) and Main Disconnect Panel (MDP) must be installed and operational prior to magnet arrival.

Note

Customer provided temporary backup water cooling is highly recommended for the Cryo Cooler Compressor Cabinet located in the lower portion of the System Cooling Cabinet (SCC). Backup temporary water cooling can be connected directly to the Cryo Cooler Compressor Cabinet via the by-pass connection located inside the SCC. Refer to Section 4-4-1 Cryo Cooler Compressor Backup Temporary Water Cooling Requirement.

Note

The SCC and system Body Coil located inside the Magnet Enclosure must be located within a maximum of 15 ft (4.6 m) vertical elevation difference and total length. Refer to Section 6, INTERCONNECT DATA, for interconnects length between SCC and Magnet Enclosure.

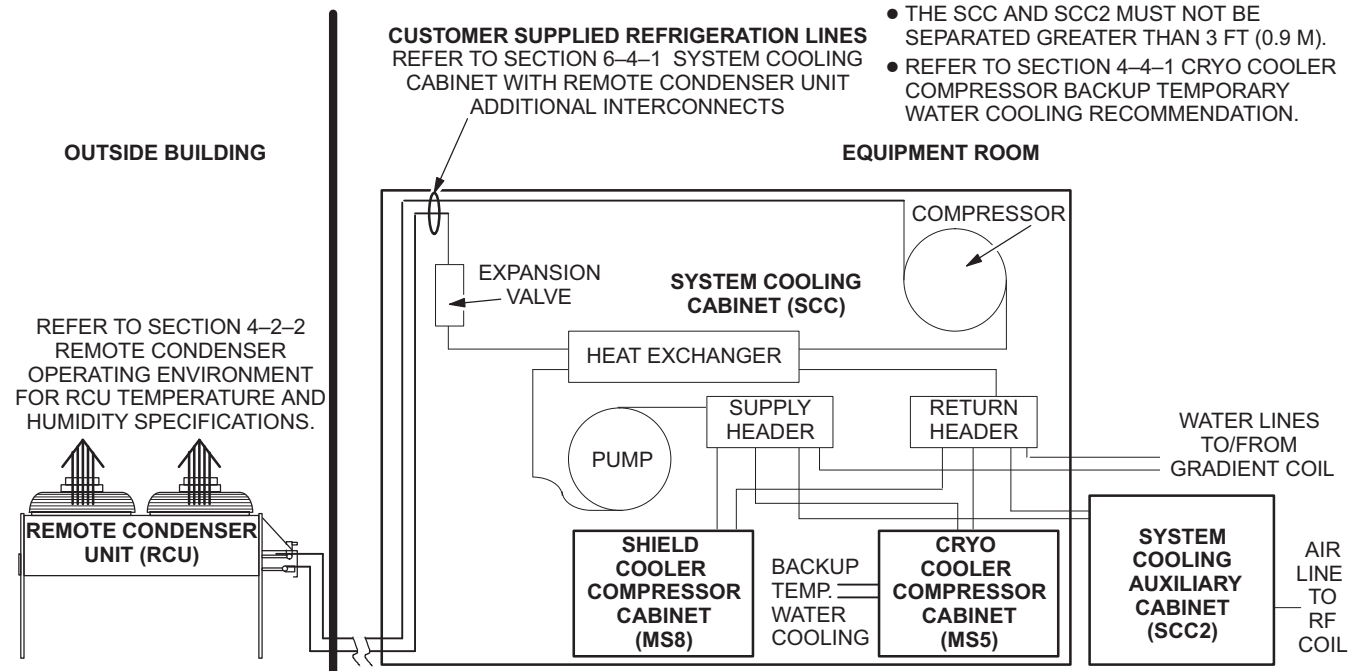
The SCC is powered from the Main Disconnect Panel (MDP) via customer supplied wiring. The SCC2 is powered from SCC via a supplied power cord. For an SCC with a RCU, the RCU is powered from the SCC via customer supplied power wiring with a Lock Out/Tag Out capable disconnect box located on the RCU. Refer to Section 6, INTERCONNECT DATA, for additional interconnects information and requirements.

2-7-1 System Cooling Cabinet Subsystem Equipment (Continued)

Air Cooled SCC with Remote Condenser Unit (RCU) has the SCC installed in the Equipment Room and the RCU installed external to the building which exhausts the majority of heat to outside air. The SCC in this configuration contains the water chiller compressor and heat exchanger which requires air cooling. See Illustration 2-3.

NOTE:

- THE SCC AND RCU MUST NOT BE SEPARATED BY A DISTANCE GREATER THAN 200 FT (61 M) OF CUSTOMER PROVIDED REFRIGERATION TUBING. THE VERTICAL SEPARATION MUST NOT EXCEED 100 FT (30.5 METER) FOR THE RCU ABOVE THE SCC AND 10 FT (3 M) FOR THE RCU BELOW THE SCC.
- THE SCC AND SCC2 MUST NOT BE SEPARATED GREATER THAN 3 FT (0.9 M).
- REFER TO SECTION 4-4-1 CRYO COOLER COMPRESSOR BACKUP TEMPORARY WATER COOLING RECOMMENDATION.



SYSTEM COOLING CABINET WITH REMOTE CONDENSER UNIT MODEL LC20M/RAC OVERVIEW SKETCH

ILLUSTRATION 2-3

2-7-1 System Cooling Cabinet Subsystem Equipment (Continued)**RCU Location and Mounting**

Provisions for a supply of ambient air to the RCU and removal of heated air from the condenser area must be taken into account when determining the location of the air cooled RCU. Failure to adhere to these essential requirements will result in higher head pressures which cause poor operation and possible eventual failure of the equipment. The RCU must not be located in the vicinity of steam, hot or fume exhausts. The RCU must be mounted with the airflow in the vertical direction and can not be mounted with the airflow in a horizontal direction.

The RCU should be mounted away from noise sensitive spaces and must have adequate support to avoid vibration and noise transmission into the building. Sound and structural consultants should be retained for recommendations.

The RCU is designed for outdoor application and may be mounted on a roof or above ground support, or a concrete slab for ground level installation.

- Roof or support mounted units should be level on steel channels or an I-beam frame to support the unit above the roof or other mounting surface. These mounts must be strong enough to support the weight of the unit. Due to the various details involved with roof or support mounting structure, Ellis and Watts International cannot determine the details for this type of mounting without the necessary engineering and site data. It is the responsibility of the installer to have the necessary resources to determine framework for roof or support mounting of the RCU.
- Ground level installation of the RCU requires a concrete pad be used. The concrete pad should be installed level and be properly supported to prevent settling. See Illustration 2-14 for concrete pad dimensions and concrete requirements. The concrete footing should meet or exceed the local code requirements.

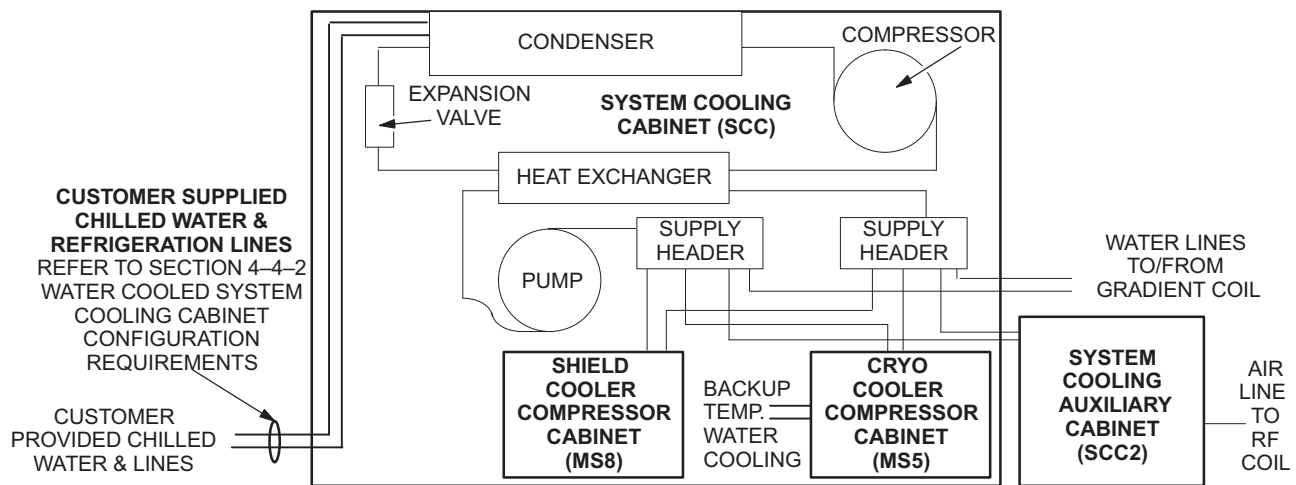
Installation of the RCU must be in accordance with local and national codes for electrical and according to standard accepted refrigeration practices i.e. proper traps in line.

2-7-1 System Cooling Cabinet Subsystem Equipment (Continued)

Water Cooled SCC with recondenser inside the cabinet is a dedicated, closed loop, liquid-to-liquid water chiller system. The water cooled SCC is installed in the Equipment Room with some heat output to room air and the majority of heat output to customer supplied chilled cooling water. Refer to Sections 4-3 AIR COOLING REQUIREMENTS and 4-4-2 Water Cooled System Cooling Cabinet Configuration Requirements. See Illustration 2-4.

NOTE:

- THE SCC AND SCC2 MUST NOT BE SEPARATED GREATER THAN 3 FT (0.9 M).
- REFER TO SECTION 4-4-1 CRYO COOLER COMPRESSOR BACKUP TEMPORARY WATER COOLING RECOMMENDATION.



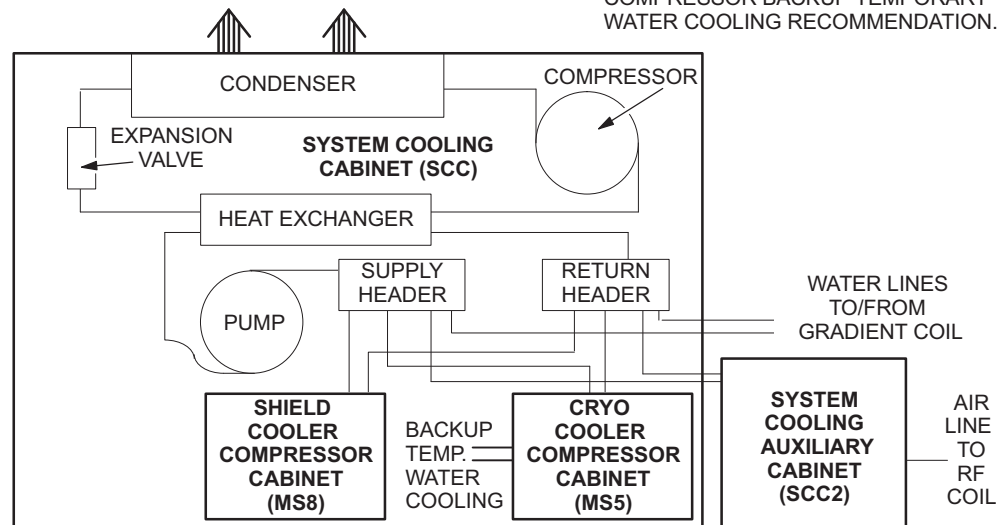
WATER COOLED SYSTEM COOLING CABINET MODEL LC20M/WC OVERVIEW SKETCH
ILLUSTRATION 2-4

2-7-1 System Cooling Cabinet Subsystem Equipment (Continued)

Air Cooled System Cooling Cabinet (SCC) with condenser in the cabinet is a dedicated, closed loop, water-to-air chiller system installed in the Equipment Room. The SCC heat output to air must be managed by the Equipment Room cooling system, refer to **Section 4-3 AIR COOLING REQUIREMENTS** for specifics. See Illustration 2-5.

NOTE:

- THE SCC AND SCC2 MUST NOT BE SEPARATED GREATER THAN 3 FT (0.9 M).
- REFER TO SECTION 4-4-1 CRYO COOLER COMPRESSOR BACKUP TEMPORARY WATER COOLING RECOMMENDATION.



AIR COOLED SYSTEM COOLING CABINET MODEL LC20M1 OVERVIEW SKETCH
ILLUSTRATION 2-5

2-7-1 System Cooling Cabinet Subsystem Equipment (Continued)**Responsibility For Installation Tasks For SCC Equipment**

The SCC subsystem equipment installation requires specific tasks to be performed by the Customer Contractors (riggers, plumbers, electrician and HVAC contractor), GE Service, and Ellis & Watts Service Provider (vendor of the SCC equipment). Refer to Table 2-5 for the list of the responsibility for the specific tasks. Refer to vendor manual for additional information concerning tasks.

TABLE 2-5
SCC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

COOLING CABINET TYPE				INSTALLATION TASKS FOR SCC EQUIPMENT See Notes 1, 2, & 3	RESPONSIBILITY		
INDOOR WITH RCU See Note 1	INDOOR WATER COOLED See Note 2	INDOOR AIR COOLED See Note 3	CUSTOMER CONTRACTORS		GE SERVICE	E&W SERVICE PROVIDER	
WHEN SCC EQUIPMENT IS AT SITE & PRIOR TO MAGNET DELIVERY							
X	X	X	Unload SCC and SCC2 and accessories from truck, uncrate and deliver to equipment installation location. (Remote Condenser Units are generally installed outside of the building.) Note: Lifting equipment such as forklift or crane must be available to perform unloading / move to final location for equipment installation.	X			
X	X	X	Install SCC2 next to SCC.	X			
X	X	X	Install SCC and mount on Vibration Isolators. (Use seismic Tie Down Brackets where required by law).	X			
X	X	X	Inflate Vibration Isolators with air or dry nitrogen.	X			
X	X	X	Install Main Disconnect Panel (MDP).	X			
X	X	X	Install customer supplied power wiring from MDP to SCC. Make final electrical connections at both ends.	X			
X			Assemble and mount Remote Condenser Unit (RCU)	X			
X			Install electric conduit and wiring between SCC and RCU. Make final electrical connections at both ends.	X			
X			Install customer supplied refrigeration piping from SCC to RCU. Make final connections; fill lines with dry nitrogen and leak test. Repair leaks. Evacuate nitrogen from lines, pull vacuum to 500 microns and fill with a holding charge of 10 pounds of R-134a refrigerant. Must be performed by qualified HVAC contractor.	X			
	X		Install water filter, water supply and water return lines from customer's chilled water supply to SCC Water Cooled Condenser.	X			
X	X	X	Install Facility Backup Chilled water supply and returns lines to SCC (Optional) (Emergency Water Back-up). Refer to Section 4-4-1 Cryo Cooler Compressor Backup Temporary Water Cooling Recommendation . Note: This feature allows service to switch over the cooling water supply from the SCC to the Facility Backup thereby allowing the 2 stage Cryo Cooler Compressor to run in the event the SCC stops working. For LC20M1/WC units – Do not use the customer's chilled water supply as the Facility Backup supply.	X			
Note 1 Refer to Ellis & Watts <i>Technical Publication 468 Pre installation/Installation/Operating LC 20M/RAC Chilled Water System for GE Signa OpenSpeed MRI Equipment</i> for additional SCC with RCU information and details. 2 Refer to Ellis & Watts <i>Technical Publication 467 Pre installation/Installation/Operating LC 20M/WC Chilled Water System for GE Signa OpenSpeed MRI Equipment</i> for additional Indoor Water Cooled SCC information and details. 3 Refer to Ellis & Watts <i>Technical Publication 460 Pre installation/Installation/Operating LC 20M Chilled Water System for GE Signa OpenSpeed MRI Equipment</i> for additional Indoor Air Cooled SCC information and details.							
(Continued)							

2-7-1 System Cooling Cabinet Subsystem Equipment (Continued)

TABLE 2-5 (Continued)

**SCC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED**

COOLING CABINET TYPE			INSTALLATION TASKS FOR SCC EQUIPMENT See Notes 1, 2, & 3	RESPONSIBILITY		
INDOOR WITH RCU See Note 1	INDOOR WATER COOLED See Note 2	INDOOR AIR COOLED See Note 3		CUSTOMER CONTRACTORS	GE SERVICE	E&W SERVICE PROVIDER
X			See APPENDIX B Prerequisites for Scheduled Start-up Or Performance Verification of Chillers For Signa Open Speed and review for completeness. Send email request for initial start up of RCU to service@elliswatts.com . Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	
X			Install power, control and water lines from SCC to SCC2, short loop the Gradient Coil water supply and return lines, start up, operate and add final refrigerant charge of R-134a and check fluid levels / operation. Must be completed 1 week prior to magnet arrival.	X		
WHEN MAGNET IS DELIVERED/INSTALLED						
X	X	X	Install Cryo Cooler and Shield Cooler compressors into position inside the lower deck of SCC.		X	
X	X	X	Install water supply and return lines from SCC to: 1) Gradient Coil* via system Penetration Panel (PP1) 2) Cryo Cooler Compressor 3) Shield Cooler Compressor Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
X	X	X	Install power, control and water return lines from SCC to SCC2. Note: SCC2 and Gradient Coil water lines must be connected or short looped in order for SCC to start. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
X	X	X	Install 2 inch air hose from SCC2 to RF Coil duct at back of Magnet. Note: If the Gradient Coil or SCC2 are not connected then must be short looped on the SCC for proper operation. SCC will not function without the four loops. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
X	X	X	Fill SCC with Propylene Glycol per service manual.		X	
X	X	X	Start up SCC, SCC2, and RCU if present and verify operation per service manual.		X	
X	X	X	See APPENDIX B Prerequisites for Scheduled Start-up Or Performance Verification of Chillers For Signa Open Speed and review for completeness. Send email request for initial performance verification to service@elliswatts.com . Note: For RCU type units this will be the 2nd visit made by the E&W Service Provider. Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	
X	X	X	Write the System ID and Magnet Monitor System ID onto the label located on the System Cooling Cabinet.		X	
X	X	X	Verify proper installation and operation with all thermal loads active per E&W Startup Checklist.			X
Note 1 Refer to Ellis & Watts <i>Technical Publication 468 Pre installation/Installation/Operating LC 20M/RAC Chilled Water System for GE Signa OpenSpeed MRI Equipment</i> for additional SCC with RCU information and details. 2 Refer to Ellis & Watts <i>Technical Publication 467 Pre installation/Installation/Operating LC 20M/WC Chilled Water System for GE Signa OpenSpeed MRI Equipment</i> for additional Indoor Water Cooled SCC information and details. 3 Refer to Ellis & Watts <i>Technical Publication 460 Pre installation/Installation/Operating LC 20M Chilled Water System for GE Signa OpenSpeed MRI Equipment</i> for additional Indoor Air Cooled SCC information and details.						

2-7-2 Pneumatic Patient Alert (PA1)

The Pneumatic Patient Alert system is a stand alone system that will allow the Patient to contact the Operator even when the intercom volume is turned down. The Control Box is to be located near the Operator Workspace. The Control Box audible and visual alarm will be activated by the patient squeeze bulb which is located on the Magnet Enclosure and connected by pneumatic tubing through the Penetration Panel to the Control Box. The Control Box should be mounted with consideration for ease of use by operator, remaining within sight of operator and within 5 ft (1.5 m) of an electrical outlet. The Control Box can be powered from an outlet on the Operator Workspace. Refer to Illustration 2-32 for Control Box mounting dimensions.

2-7-3 SPT Phantom Set Shipping/Storage Cabinet

System Performance Test provides the customer and GE Service with a means to quickly verify whether critical parameters affecting image quality are within specifications. The test uses a set of phantoms and a nesting plate for proper positioning of the phantoms on the Patient Table. The phantom set and nesting plate are provided in a cabinet which protects the pieces during shipment and storage at site. The cabinet is not magnetic therefore it can be stored inside the Magnet Room if so desired and moved to the Patient Table for ease of positioning the phantoms. See Illustration 2-34 for dimensions information.

2-7-4 Magnet Monitor Equipment

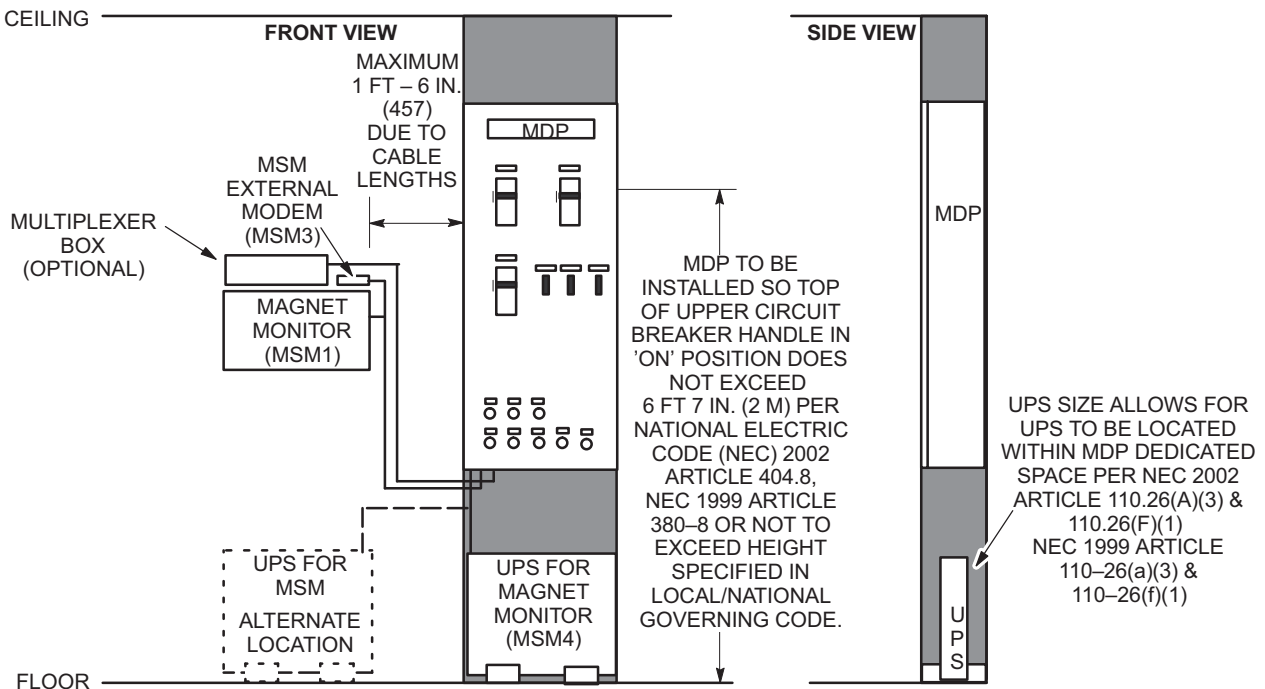
The Magnet Monitor (MSM1) performs the functions of a cryogen meter and magnet pressure control with readout display capability on the unit. Also the Magnet Monitor with its modem allows for remote monitoring during system warranty period or available as part of a GE Service Contract. A dedicated direct-distance-dialing voice-grade telephone line with access located near the Magnet Monitor in the Equipment Room for remote monitoring is required, refer to **Section 2-7-5 System Monitoring & Support Connectivity Requirements** for additional information. The Signa OpenSpeed includes a Remote Alarm Box to be located on or near the Operator Workspace that has visual and audible alerts if the Magnet Monitor detects a major alarm condition.

Magnet Monitor remote monitoring operation is critical to maximize proper uninterrupted 0.7T magnet operation. A small Uninterruptible Power System (UPS) is provided with the Magnet Monitor for the 0.7T Magnet. In the event of facility power outage, the UPS will maintain Magnet Monitor operation for sufficient time to communicate status via the remote monitoring.

The Magnet Monitor and its UPS are powered from the Main Disconnect Panel (MDP) and must be mounted near the MDP, see Illustration 2-6 The Magnet Monitor should be mounted approximately 60 in. (1524 mm) above the floor in the Equipment Room. Refer to **Section 6 – INTERCONNECT DATA** for details of Magnet Monitor, UPS, and system interconnects.

NOTE:

SPACE ABOUT ELECTRICAL EQUIPMENT PER NATIONAL ELECTRICAL CODE (NEC) 2002 ARTICLE 110.26 OR NEC 1999 ARTICLE 110-26.



MAGNET MONITOR (MSM) & ASSOCIATED EQUIPMENT, UPS FOR MAGNET MONITOR, AND MDP LOCATIONS

ILLUSTRATION 2-6

2-7-5 System Monitoring & Support Connectivity Requirements

One of the system monitoring and support connectivity configurations listed in Table 2-6 must be provided for system installation and serviceability purposes. The network connection (if selected configuration) and telephone lines are to be provided and paid for by the customer.

Note

Telephone line and network connection (if planned configuration) requirements listed in Table 2-6 must be available and functioning prior to magnet delivery and for duration of installation for communication with Magnet Monitor equipment.

TABLE 2-6
SYSTEM MONITORING & SUPPORT CONNECTIVITY REQUIREMENTS

CONFIGURATION	PHONE LINE	USE/LOCATION
Network connection & telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	Two Network connection with a static IP Addresses See Note 1	One access located near the Operator Workspace (OW) in the Control Room.
		One access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. This Ethernet connection must not loose power when the MR system is shutdown.
Multiple telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Operator Workspace (OW) in the Control Room. See Note 2 & 3
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. See Note 2 & 3
Note 1 For North America & South America: One Internet accesible Virtual Private Network Connection with a static IP Address is required. For Europe: One ISDN Network Connection with a static IP Address is required.		
2 A dedicated direct-distance-dialing <u>voice-grade</u> telephone line can be shared for Operator Workspace (OW) and Magnet Monitor (MSM) requirement through the use of a multiplexer box. The following multiplexer boxes are available for customer purchase. 46-328475P1 4 Line Phone Multiplexer box; 115 VAC input power 46-328475P3 4 Line Phone Multiplexer box; 220 VAC input power If the customer chooses not to purchase the multiplexer box then the customer must provide an additional line for each requirement as stated in this table.		
3 If a Multiplexer Box is used then the Magnet Monitor MUST be Channel 1 to allow for call out after a power outage.		

2-7-6 Step-Up Transformers for 200/208 V Sites

A step-up transformer is required if the site input voltage for the system is 200V or 208V. The system Main Disconnect Panel (MDP) & System Cooling Cabinet (SCC) configurations require high voltage for input voltage. For transformer options available from GE refer to Section 1-2 BASIC SYSTEM, Table 1-6.

2-8 ARCHITECTURAL REMINDERS

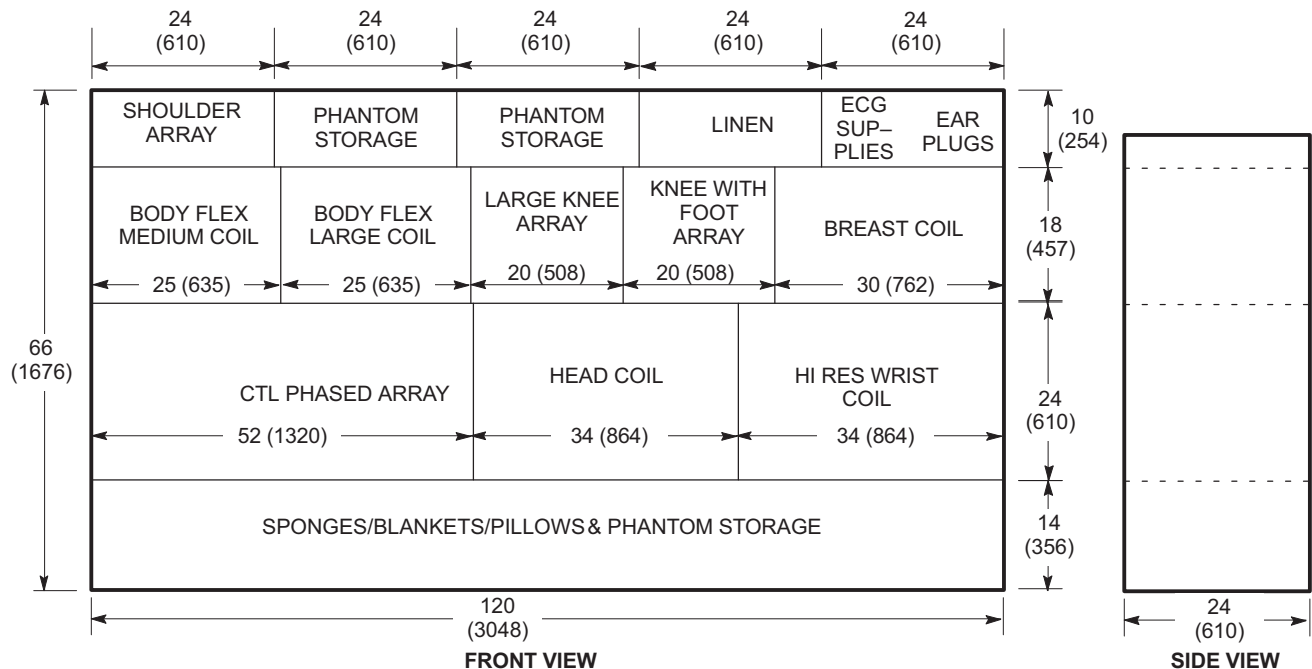
1. Pay attention to isogauss limits, not only for placement of equipment in rooms, but also for isogauss limits with respect to outside environment.
2. The customer is responsible for establishing protocols to warn persons with cardiac pacemakers, neurostimulators, and biostimulation devices of the potential danger of entering magnetic fields greater than 5 gauss (exclusion zone).
3. Due to the periodic cryogen servicing of the magnet, consideration must be given to the delivery route of the cryogens to the Magnet Room. The service route should be level and therefore steps or steep ramps must be avoided. Maximum acceptable incline along dewar delivery route is 1:12 (5°).
- 3A. Cryogen dewars must not be stored within the Magnet Room due to the safety issues of seismic considerations, spillage effects, fire hazards and explosive effects of compressed gas. Also the magnetic field inhomogeneity is affected by the physical shape of the non-magnetic dewars. If the magnet is shimmed with no dewars in the Magnet Room, the magnetic field inhomogeneity will slightly change once dewars are moved into the room (or vice versa). All dewars must be stored outside of the Magnet Room and more than 10 ft (3.05 m) from isocenter of the magnet in all directions.
- 3B. Means must be provided to secure gas cylinders used for cryogen transfills in an upright position using a removable chain or strap. This is to prevent the cylinders from falling, which may cause injury or damage.
4. If elevators are to be used along cryogen delivery route, verify that elevator dimensions and weight capacity is sufficient to handle the cryogen dewars. Also, elevator must be dedicated with restricted access during cryogen transport (will not allow stops between initial start and final floor destination).
5. The operator seated at the Operator Workspace should have an unobstructed view of the patient on the Patient Table. A typical height of 42 inches (1067 mm) to the bottom of the Magnet Room viewing window should allow viewing of the patient on the Patient Table.
6. It is recommended that the Magnet Room viewing window be of fine mesh screening material (as opposed to a “honeycomb-type pattern”) for better visibility of the patient from the Operator Workspace.
7. Operators in Magnet Room must have easy access to the scan control switches located on both sides of front top panel of the magnet enclosure.
8. A patient preparation/emergency area should be located near the Magnet Room and direct patient access must be available from the Magnet Room to a patient preparation/emergency area.
9. Customer provided and paid for telephone lines must be supplied for system installation and serviceability purposes per **Section 2-7-5 System Monitoring & Support Connectivity Requirements**.
10. A lockable storage cabinet can be provided and maintained by GE Medical Systems Service for storage of GE Medical Systems service documentation/tools. Cabinet to be approximately 36 in. (914 mm) wide, 18 in. (457 mm) depth, and 72 in. (1829 mm) high.

2-8 ARCHITECTURAL REMINDERS (Continued)

11. Corrosive chemicals must not be stored or used in the Equipment Room. These include chemicals used for film processor storage tanks, processor chemical recovery systems, etc. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control area.
12. Storage space for system accessories and supplies should be planned for and included in room layout drawings. Illustration 2-7 shows one suggestion for a shelf/cabinet arrangement developed by several MR Application Specialists and MR System Operators.

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



SUGGESTION FOR ACCESSORIES & SUPPLIES STORAGE SHELF/CABINET
ILLUSTRATION 2-7

2-9 FLOOR LOADING AND WEIGHTS

This section contains loading considerations for the MR system. Listed in Table 2-7 are the weights, floor loading, and normal mounting methods for MR components.

TABLE 2-7
FLOOR LOADING

COMPONENT	NET WT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
FOR SLAB ON GRADE SITE: Magnet, Enclosure, and Coils	See Notes 1 & 4 & Refer to Section 2-9-1	81.5 x 81.5 x 95.8 (2070 x 2070 x 2434)	See Illustration 7-15 through 7-19	Magnet & Table Rail Assembly anchor/bolted to floor with supplied materials.
FOR SUSPENSION SITE: Magnet, Magnet Mounting Plate, Enclosure, and Coils	See Notes 2 & 4 & Refer to Section 2-9-1	81.5 x 81.5 x 95.8 (2070 x 2070 x 2434)	See Illustration 7-15 through 7-19	Magnet & Table Rail Assembly bolted to magnet mounting plate. Plate is installed with epoxy & anchor/bolted to floor with supplied materials.
Patient Swing Table (See Note 5)	1300 (591)	36 x 97 x 36 (914 x 2464 x 914)	See Illustrations 2-8 & 2-9	Magnet end of Patient Swing Table connects to arc rail attached to magnet mounting plate.
Air Cooled System Cooling Cabinet utilize with RCU (See Note 6 & 7)	1720 (782)	65 x 32.5 x 75.44 (1651 x 826 x 1916)	Rectangular base 65 x 32.5 (1651 x 816). Six vibration isolators 3.88 (98.5) dia. See Illustration 2-12.	Casters for location. Set on floor on vibration isolators. Tie down brackets also used for seismic zones.
Remote Condenser Unit (RCU)	470 (214)	48 x 75 x 44 (1219 x 1905 x 1118)	See to Illustration 2-13 for mounting pattern.	Installed on vibration isolation pads.
Water Cooled System Cooling Cabinet (See Note 6 & 7)	1750 (795)	65 x 32.5 x 75.44 (1651 x 826 x 1916)	Rectangular base 65 x 32.5 (1651 x 816). Six vibration isolators 3.88 (98.5) dia. See Illustration 2-16.	Casters for location. Set on floor on vibration isolators. Tie down brackets also used for seismic zones.
Air Cooled System Cooling Cabinet (See Note 6 & 7)	1870 (850)	65 x 32.5 x 75.44 (1651 x 826 x 1916)	Rectangular base 65 x 32.5 (1651 x 816). Six vibration isolators 3.88 (98.5) dia. See Illustration 2-16.	Casters for location. Set on floor on vibration isolators. Tie down brackets also used for seismic zones.
System Cooling Auxiliary Cabinet (See Note 8)	550 (250)	34.12 x 40.5 x 42.12 (869 x 1029 x 1070)	Rectangular base 31.13 x 35.12 (790 x 829).	Set on floor on casters.
Note 1 Weight of magnet, enclosure, RF/Gradient Body Coil, and cryogenics is 20,180 lbs (9173 kg). 2 Weight of magnet with magnet mounting plate, enclosure, RF/Gradient Body Coil, and cryogenics is 23,180 lbs (10,536 kg). 3 Consult a structural engineer on method of calculating proper weight/unit area for floor loading. 4 Refer to Section 2-4 MINIMUM DOOR/HALLWAY SIZES for Gradient Coil Assembly replacement weight and dimension requirements. 5 Patient Table weight includes 500 lbs (227 kg) patient. 6 The System Cooling Cabinet (SCC) provides water cooling for the Cryo Cooler Compressor Cabinet, the Shield Cooler Compressor Cabinet, Gradient Coils, and System Cooling Auxiliary Cabinet. Both the Cryo Cooler Compressor Cabinet & the Shield Cooler Compressor Cabinet are physically located inside the lower portion of the System Cooling Cabinet. 7 System Cooling Cabinet is available in 3 configuration options (must choose one): • Air Cooled SCC utilized with outdoor Remote Condenser Unit (RCU) • Indoor Water Cooled SCC • Indoor Air Cooled SCC 8 The System Cooling Cabinet and the System Cooling Auxiliary Cabinet must be located side-by-side.				
(Continued)				

2-9 FLOOR LOADING AND WEIGHTS (Continued)

TABLE 2-7 (Continued)
FLOOR LOADING

COMPONENT	NET WT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
UPS for Magnet Monitor	34 (15)	19.4 x 3.5 x 17 (494 x 89 x 432)	See Illustration 2-23	Set on floor on 2 mounting brackets.
Power Cabinet	1200 (545)	23.312 x 40 x 76.13 (592 x 1016 x 1933)	Rectangular base 22 x 33 (584 x 838). Four leveling pads each 1.5 (38) dia.	Casters for location. Set on floor on leveling pads.
System Cabinet	494 (225)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four leveling pads each 1.5 (38) dia.	Casters for location. Set on floor on four leveling pads.
Operator Workspace Table with LCD Color Monitor (See Note 9)	175 (80)	54 x 43 x 52 (1372 x 1092 x 1321)	See Illustration 2-26.	Set on floor: Table on leveling pads & Cabinet rest on casters. Anchor Table to floor per Section 2-9-3.
Patient Comfort Compressor	45 (20)	19.68 x 9.60 x 17.16 (500 x 244 x 436)	Wall hanger key holes on back of unit or four rubber feet on bottom. See Illustration 2-21.	Hang on wall, set on floor or other horizontal surface.
Operator Workspace Cabinet (See Note 9)	192 (87)	18.5 x 29 x 26 (470 x 737 x 660)	See Illustration 2-27.	Set on floor and rest on casters.
R4500AS – 150 KVA 208–480Y277 Volt, 60 Hz GE transformer *	875 (397)	35 x 23 x 46 (885 x 589 x 1168)	See Illustration 2-38.	Set on floor
R4500BD – 150 KVA 200–400Y230 Volt, 50/60 Hz GE transformer *	1160 (526.3)	38 x 28 x 48 (972 x 722 x 1219)	See Illustration 2-39.	Set on floor
500 Liter Helium Dewar**	812 (369)	42 dia. x 67.5 high (1067 x 1715)	Mobile.	Four casters.
250 Liter Helium Dewar**	519 (236)	36 dia. x 64.0 high (914 x 1626)	Mobile.	Four casters.
Note * Optional Equipment. ** Dewar specifications may vary. Check with cryogen supplier for exact weight and dimensions. 9 The Operator Workspace Cabinet minimum area location is under the Workspace Table, see in Illustration 2-26. An alternate location is possible with the Operator Workspace Cabinet located against the right side of the Workspace Table.				

2-9-1 Magnet Loading Considerations

In addition to the weight of the riggers equipment, special consideration must be given to the weight of the magnet along the delivery route. Refer to **Section 9 SHIPPING AND DELIVERY DATA**, for the shipping weight of the magnet (i.e. magnet shipped with liquid cryogen and RF/Gradient coil inside magnet bore and without an enclosure). Structural reinforcement may be required along the magnet delivery route. It is required that a structural engineering analysis be performed on the Magnet Room floor and delivery route to determine its load bearing capacity.

Refer to **Section 7-6-4 FLOORS** for magnet room levelness requirements and **Section 7-7-4 Physical Requirements** for magnet mounting plate floor requirements.

2-9-2 Anchoring And Seismic Considerations

The center of gravity for MR system components are given for use in seismic calculations. If the MR cabinets are required by code to be anchored, refer to seismic drawings available on request from your local GEMS Installation Specialist.

Slab On Grade sites, excepting those in seismic zones, require the magnet to be mounted to the concrete foundation. The magnet support ring is bolted to the concrete foundation through foam isolation pads. The lower coldhead is bolted to the concrete floor through a bracket to minimize image artifacts generated by coldhead vibration. Refer to **Section 7-7-3 Slab On Grade Sites Physical Requirements** particularly Table 7-5 and Illustrations 7-12 through 7-14, for description of the site physical requirements.

For Seismic Zones and Suspension Foundation sites the 0.7T Magnet requires the magnet mounting plate be installed into the Magnet Room floor so the top of the plate will be flush with the concrete subfloor and will allow installation of the room finished floor such that the finished floor elevation is 0.25 ± 0.04 inches (6.4 ± 1.0 mm) higher than the top of the mounting plate, refer to Illustrations 7-15 through 7-19. The plate must be recessed into an existing floor. The magnet mounting plate must be utilized as the RF Shield under the Magnet and the magnet mounting plate must be mounted to the concrete. In cases where the RF Shield is located well below the magnet, i.e. in a combined RF and Magnetic shield design, then the magnet mounting plate will not be utilized as part of the RF Shield. Refer to **Section 7-7 MAGNET MOUNTING REQUIREMENTS INSIDE RF SHIELDED ROOM** for magnet mounting information.

Note

It is the customer's responsibility to coordinate and install magnet mounting plate connection and mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems.

It is the responsibility of the customer to obtain any and all approvals necessary for the construction of equipment support and seismic anchoring.

2-9-3 Operator Workspace Mounting Requirements

The Operator Workspace Table sets on the floor with the Workspace Cabinet positioned under the Table towards the right side, see Illustration 2-26. Note, the Workspace Table must be bolted to the floor for safe use of the table and equipment positioned on the table.

2-10 COMPONENT DIMENSIONS

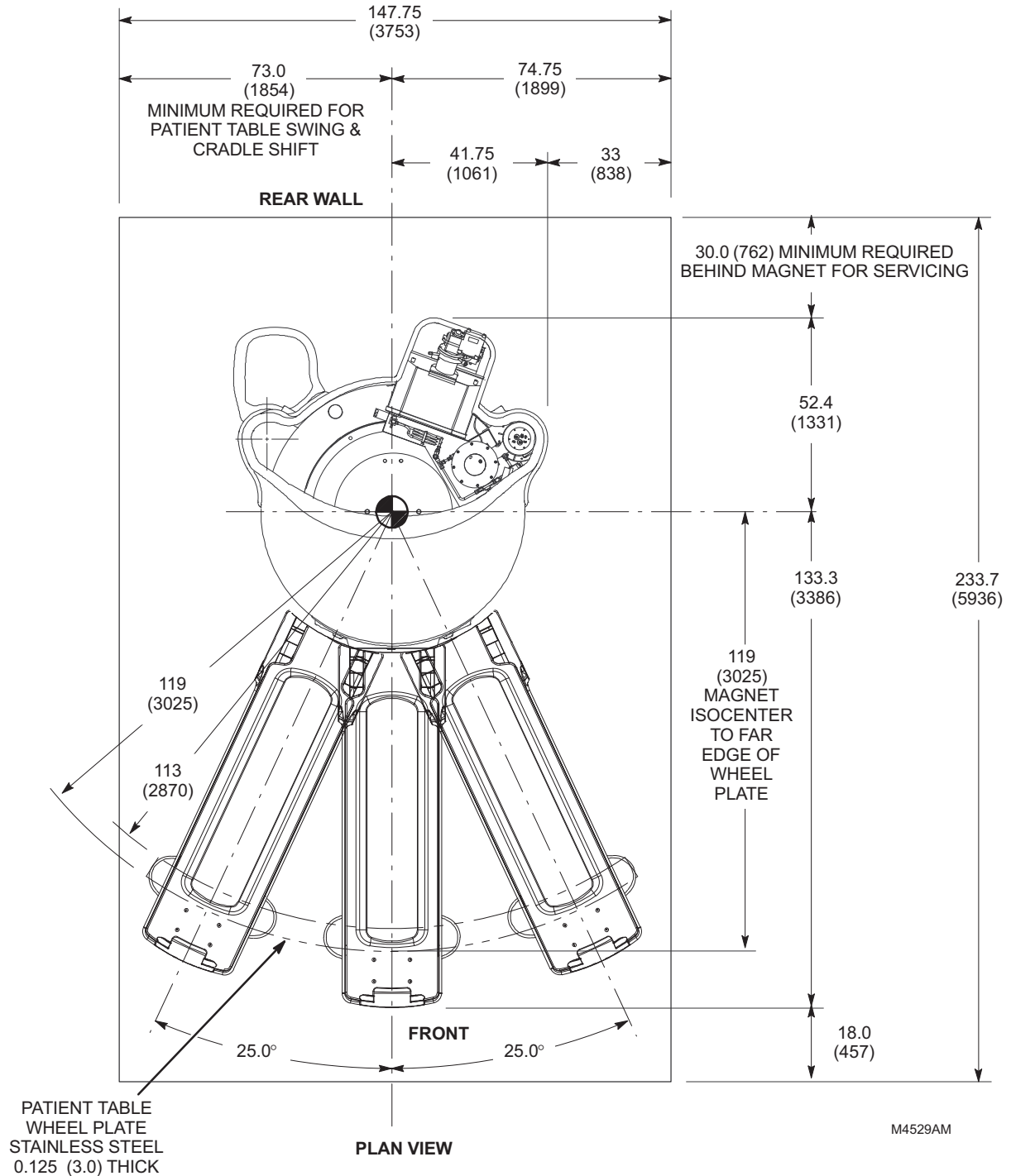
To assist in completing your room layout, refer to Table 2-8 for list of component Illustrations.

TABLE 2-8
MR SYSTEM COMPONENT ILLUSTRATIONS LIST

ILLUSTRATION NAME	ILLUSTRATION NUMBER
0.7T MAGNET, ENCLOSURE, & PATIENT TABLE (MINIMUM SERVICE AREA)	2-8
0.7T MAGNET, ENCLOSURE, & PATIENT TABLE (FRONT AND REAR VIEWS)	2-9
0.7T MAGNET, ENCLOSURE, & PATIENT TABLE CABLE ACCESS	2-10
MAIN DISCONNECT PANEL (MDP)	2-11
AIR COOLED SYSTEM COOLING CABINET [LC-20M/RAC] (SCC) UTILIZED WITH RCU	2-12
REMOTE CONDENSER UNIT [LC-20M/RAC] (RCU) FOR SCC	2-13
REMOTE CONDENSER UNIT [LC-20M/RAC] (RCU) GROUND LEVEL MOUNTING	2-14
WATER COOLED SYSTEM COOLING CABINET [LC-20M/WC] (SCC)	2-15
AIR COOLED SYSTEM COOLING CABINET [MODEL LC-20M] (SCC)	2-16
SYSTEM COOLING CABINET (SCC) MOUNTING	2-17
POWER CABINET (MR3)	2-19
SYSTEM CONTROL CABINET (MR2)	2-20
PATIENT COOLING COMPRESSOR	2-21
MAGNET MONITOR (MSM)	2-22
UPS FOR MAGNET MONITOR	2-23
PENETRATION PANEL (PP1)	2-24
PENETRATION PANEL COVER	2-25
OPERATOR WORKSPACE (OW1)	2-26
OPERATOR WORKSPACE CABINET (OW1 A2)	2-27
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR	2-28
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER	2-29
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – KEYBOARD	2-30
MAGNET MONITOR REMOTE ALARM BOX (MSM2) POSITIONED ON OW TABLE TOP	2-31
PNEUMATIC PATIENT ALERT CONTROL BOX (PA1)	2-32
MAGNET RUNDOWN UNIT (MS4)	2-33
SPT PHANTOM SET SHIPPING/STORAGE CART	2-34
DC LIGHTING CONTROLLER – GE FACILITY OPTION	2-35
OXYGEN MONITOR (OM1) – OPTIONAL	2-36
REMOTE OXYGEN SENSOR MODULE (OM3) – OPTIONAL	2-37
150 KVA 208–480Y277 VOLT, 60 Hz GE TRANSFORMER – E4500AS	2-38
150 KVA 200–400Y230 VOLT, 50/60 Hz GE TRANSFORMER – R4500BD	2-39

NOTE:

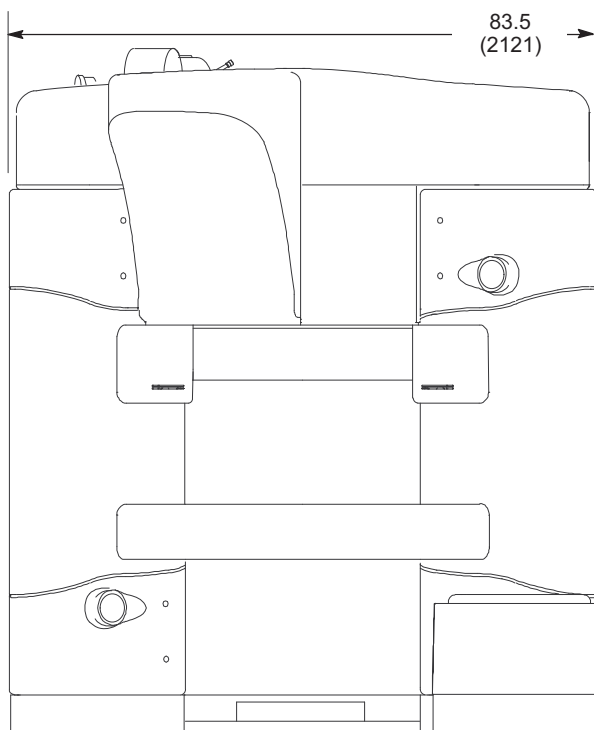
- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



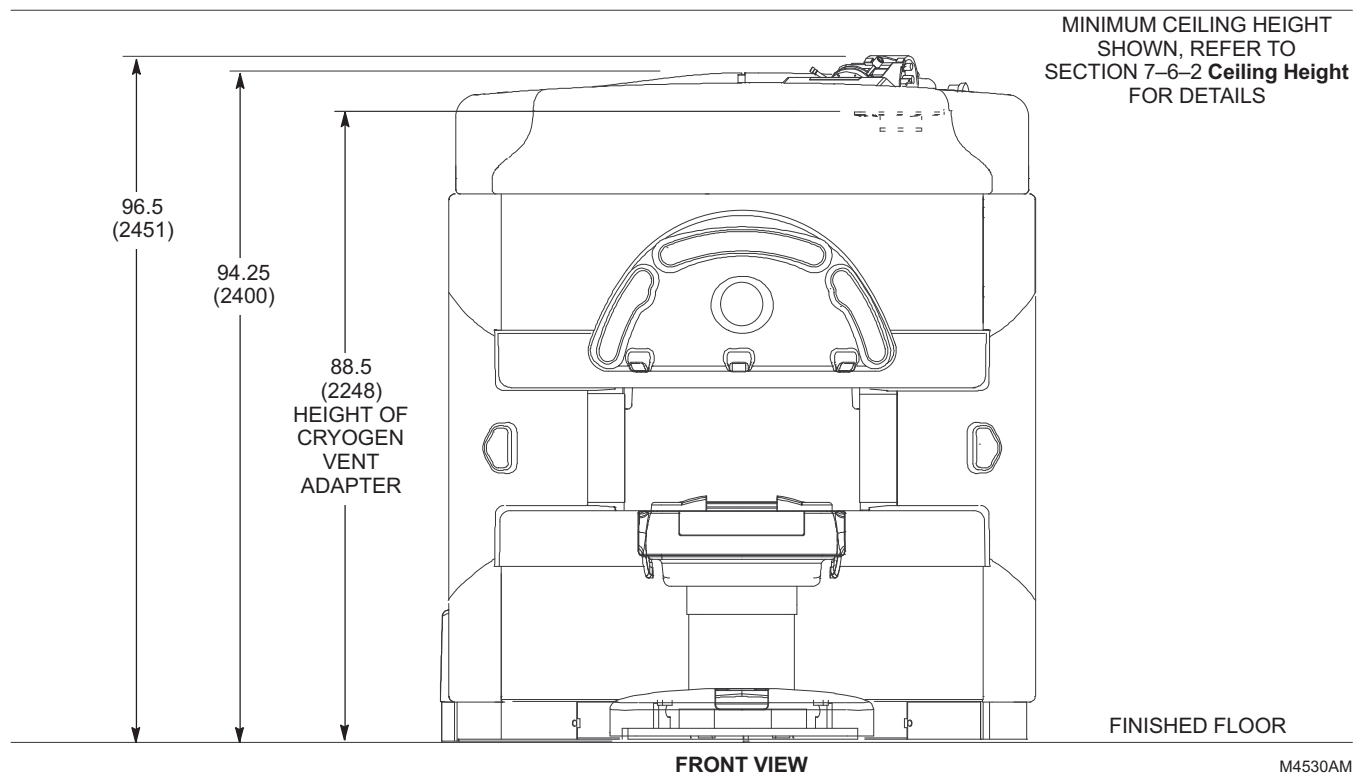
0.7T MAGNET, ENCLOSURE, & PATIENT TABLE (MINIMUM SERVICE AREA)
ILLUSTRATION 2-8

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.



REAR VIEW



FRONT VIEW

0.7T MAGNET, ENCLOSURE, & PATIENT TABLE (FRONT AND REAR VIEWS)

ILLUSTRATION 2-9

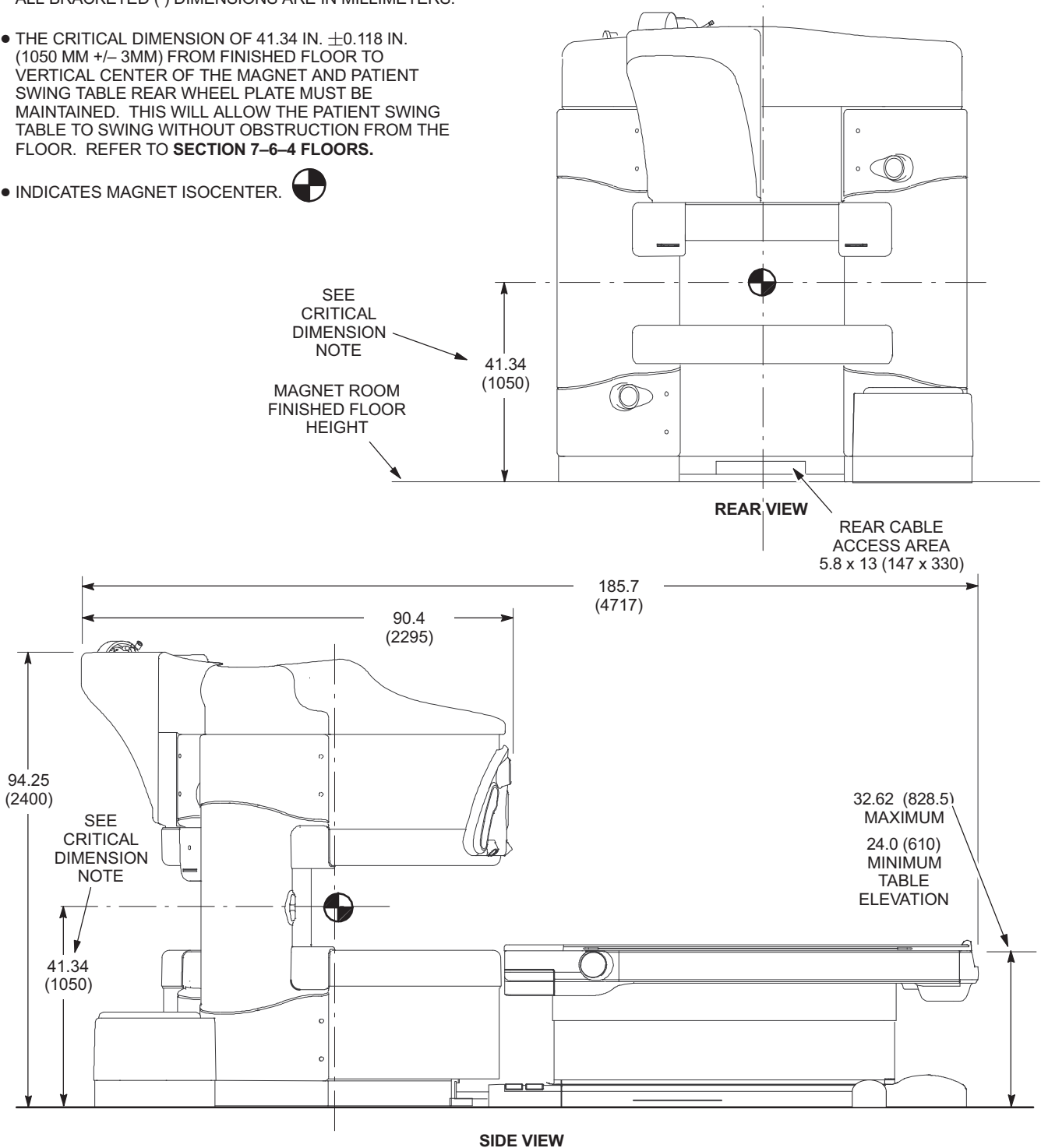
M4530AM

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

- THE CRITICAL DIMENSION OF 41.34 IN. ± 0.118 IN. (1050 MM ± 3 MM) FROM FINISHED FLOOR TO VERTICAL CENTER OF THE MAGNET AND PATIENT SWING TABLE REAR WHEEL PLATE MUST BE MAINTAINED. THIS WILL ALLOW THE PATIENT SWING TABLE TO SWING WITHOUT OBSTRUCTION FROM THE FLOOR. REFER TO **SECTION 7-6-4 FLOORS**.

- INDICATES MAGNET ISOCENTER. 



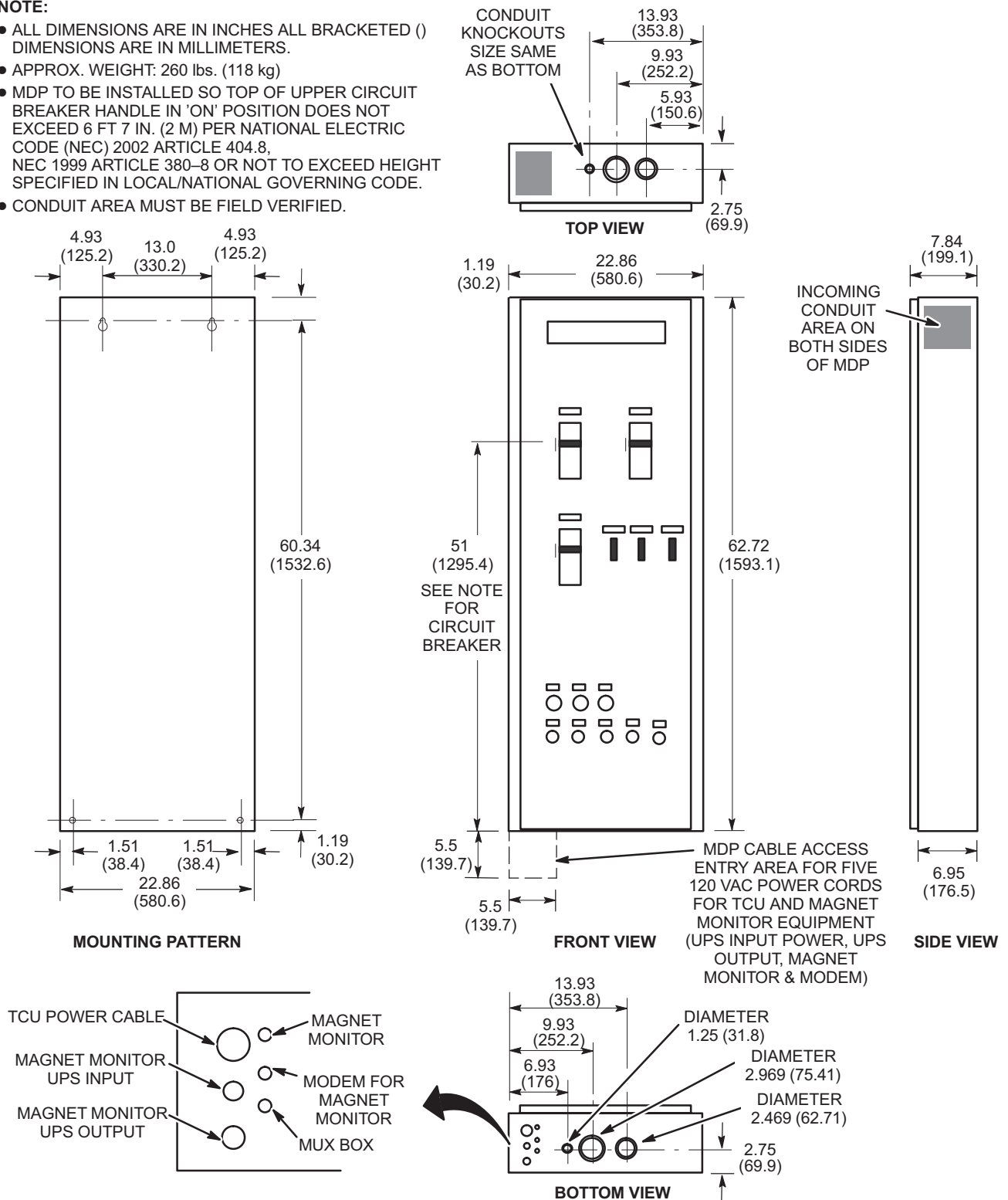
M4531AM

0.7T MAGNET, ENCLOSURE, & PATIENT TABLE CABLE ACCESS

ILLUSTRATION 2-10

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 260 lbs. (118 kg)
- MDP TO BE INSTALLED SO TOP OF UPPER CIRCUIT BREAKER HANDLE IN 'ON' POSITION DOES NOT EXCEED 6 FT 7 IN. (2 M) PER NATIONAL ELECTRIC CODE (NEC) 2002 ARTICLE 404.8, NEC 1999 ARTICLE 380-8 OR NOT TO EXCEED HEIGHT SPECIFIED IN LOCAL/NATIONAL GOVERNING CODE.
- CONDUIT AREA MUST BE FIELD VERIFIED.

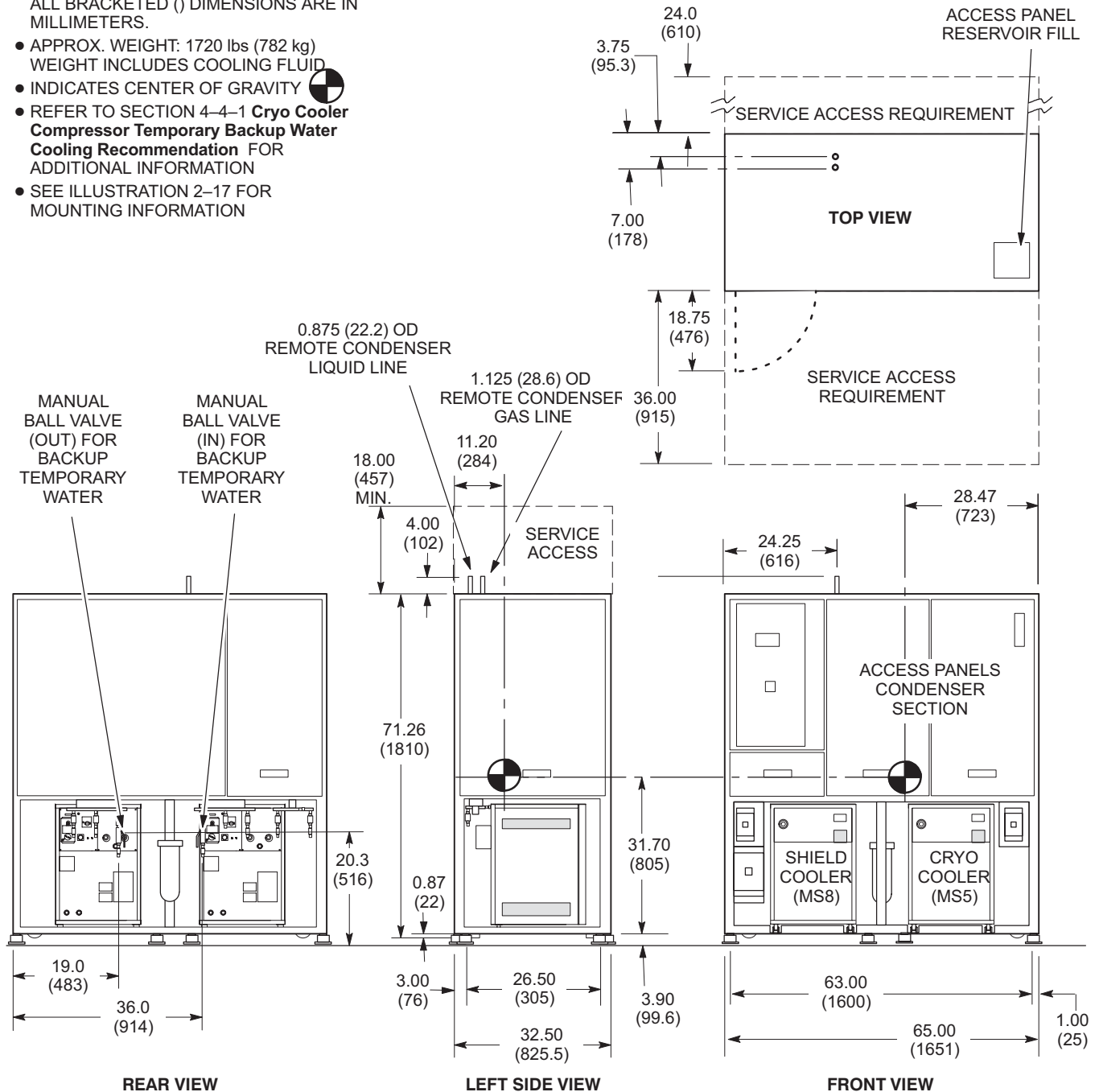


MAIN DISCONNECT PANEL (MDP)

ILLUSTRATION 2-11

NOTE:

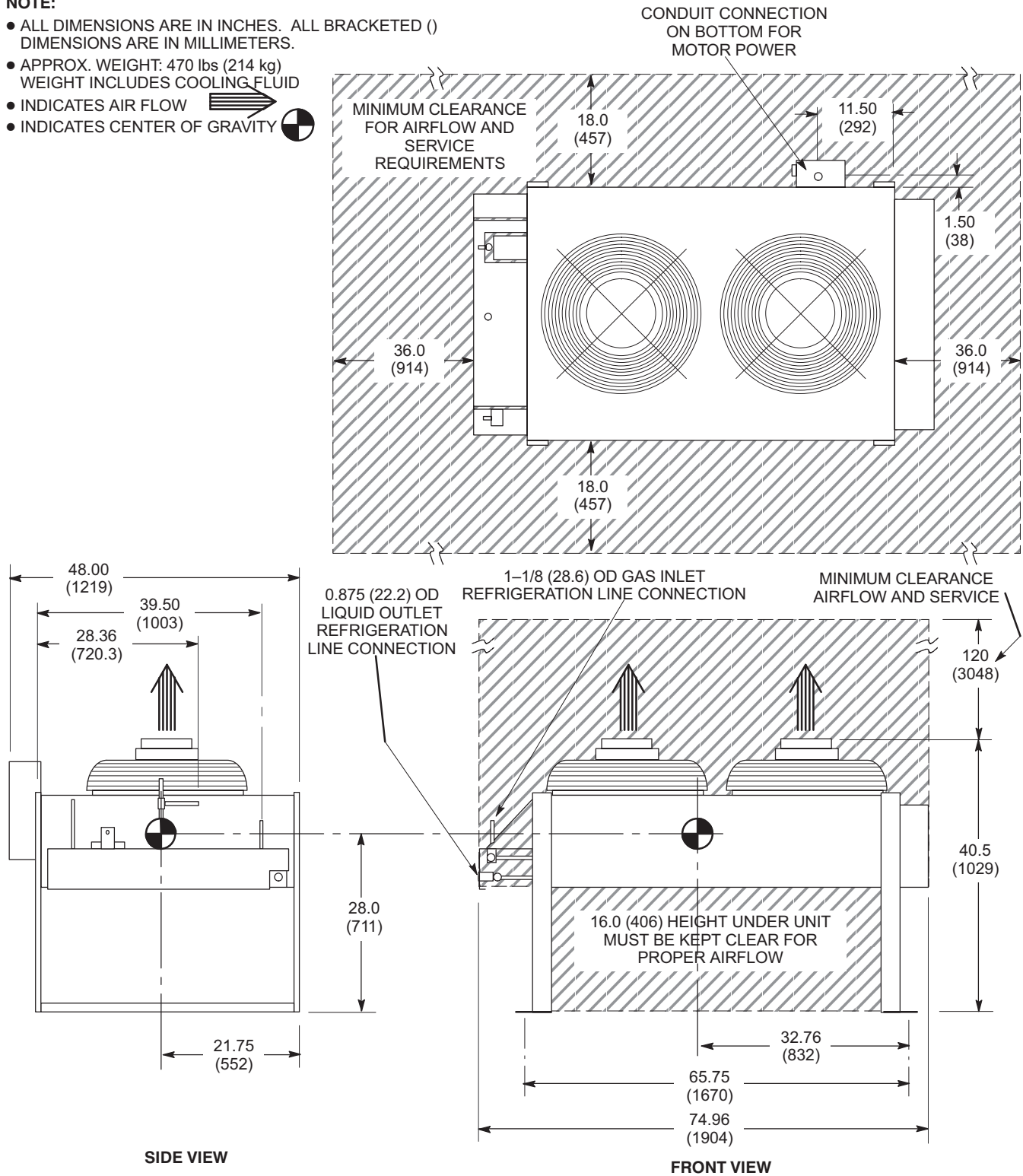
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1720 lbs (782 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES CENTER OF GRAVITY 
- REFER TO SECTION 4-4-1 **Cryo Cooler Compressor Temporary Backup Water Cooling Recommendation** FOR ADDITIONAL INFORMATION
- SEE ILLUSTRATION 2-17 FOR MOUNTING INFORMATION



AIR COOLED SYSTEM COOLING CABINET [MODEL LC20M1/RAC] (SCC) UTILIZED WITH RCU
ILLUSTRATION 2-12

NOTE:

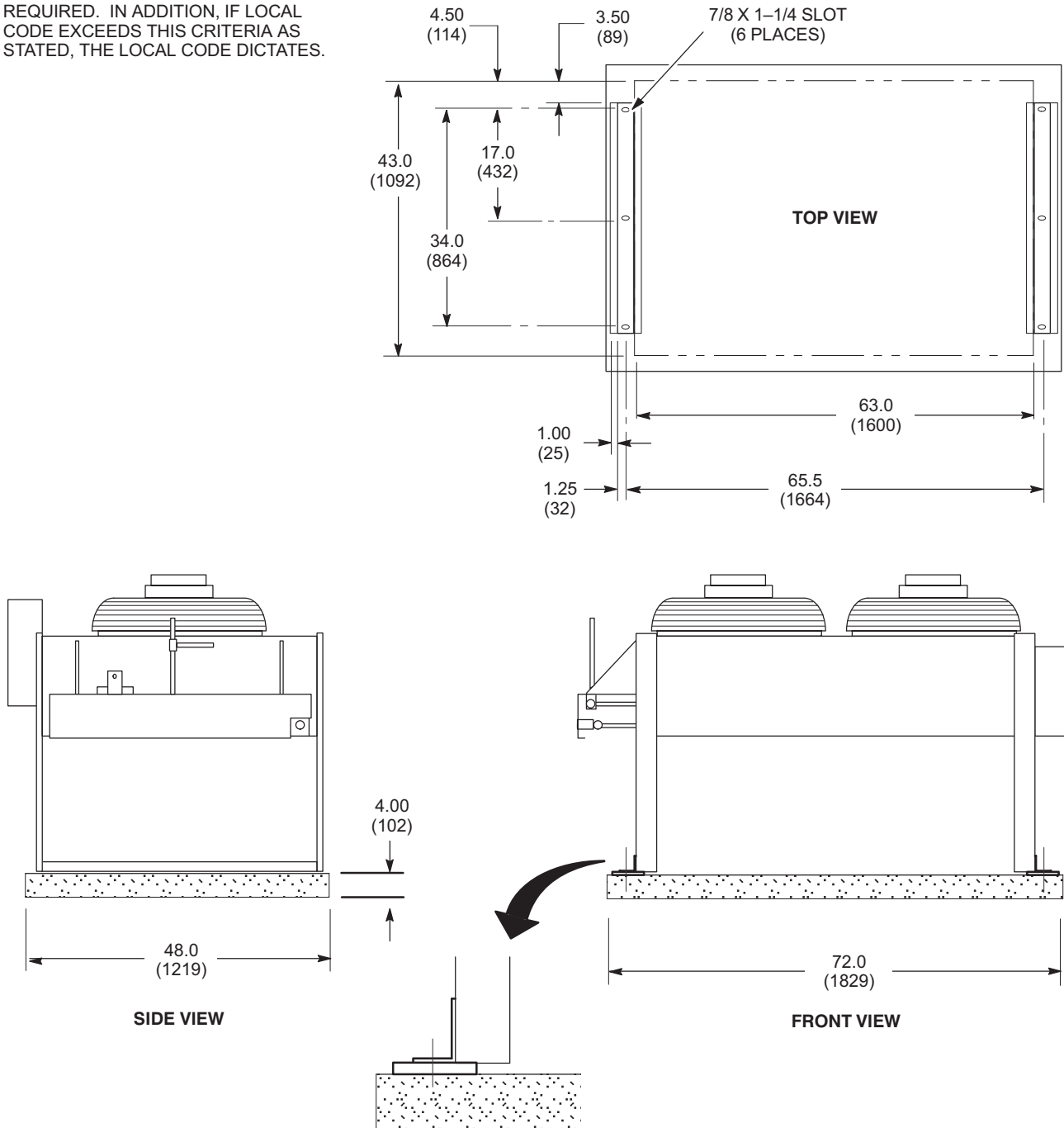
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 470 lbs (214 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 



REMOTE CONDENSER UNIT [MODEL LC20M1/RAC] (RCU) FOR SCC
ILLUSTRATION 2-13

NOTE:

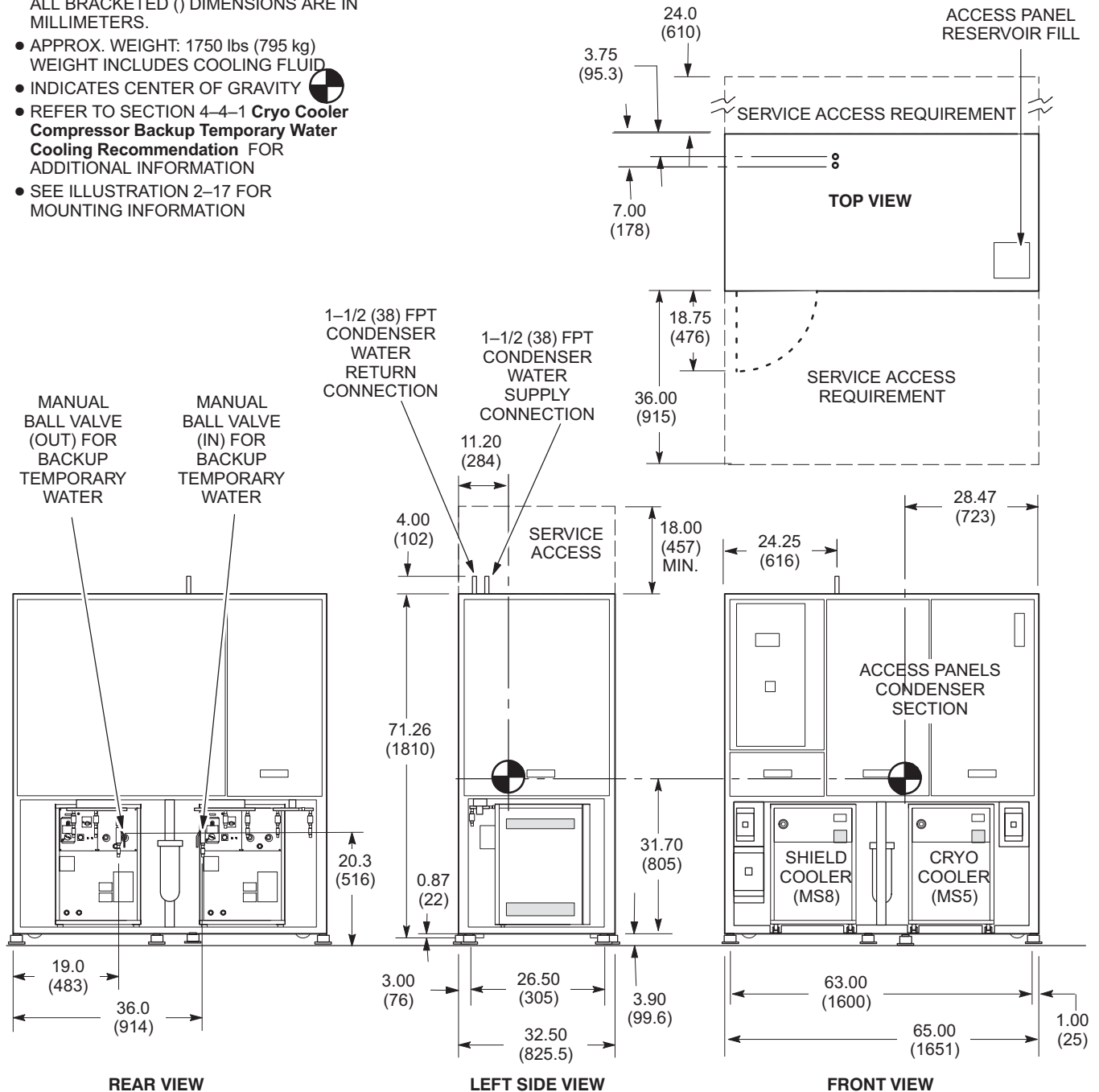
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MINIMUM 4 INCH (102 MM) DEEP CONCRETE PAD OF 2500 PSI CONCRETE REQUIRED. IN ADDITION, IF LOCAL CODE EXCEEDS THIS CRITERIA AS STATED, THE LOCAL CODE DICTATES.



REMOTE CONDENSER UNIT [MODEL LC20M1/RAC] (RCU) GROUND LEVEL MOUNTING
ILLUSTRATION 2-14

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1750 lbs (795 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES CENTER OF GRAVITY 
- REFER TO SECTION 4-4-1 **Cryo Cooler Compressor Backup Temporary Water Cooling Recommendation** FOR ADDITIONAL INFORMATION
- SEE ILLUSTRATION 2-17 FOR MOUNTING INFORMATION

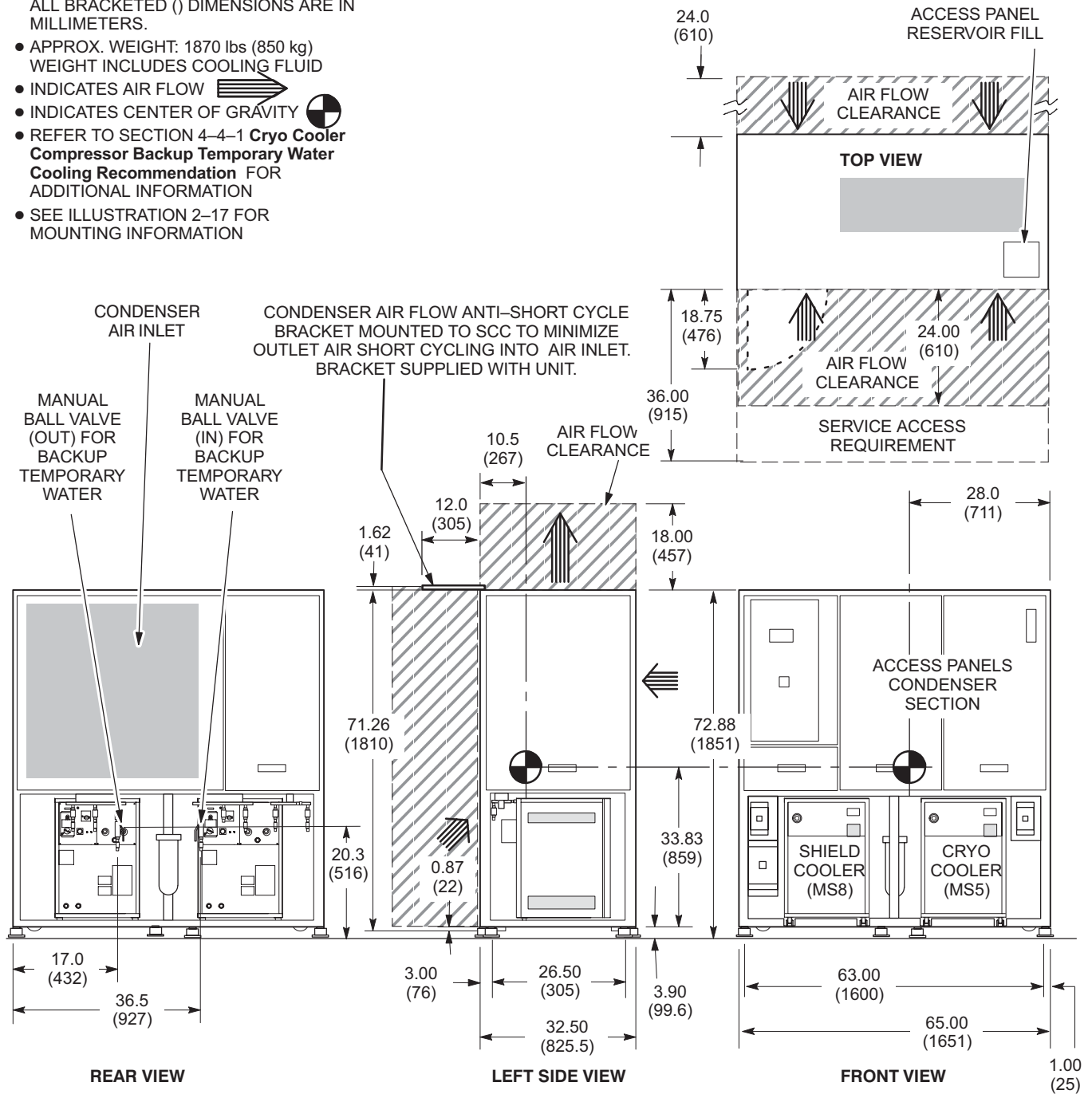


WATER COOLED SYSTEM COOLING CABINET [MODEL LC20M1/WC] (SCC)

ILLUSTRATION 2-15

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1870 lbs (850 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 
- REFER TO SECTION 4-4-1 **Cryo Cooler Compressor Backup Temporary Water Cooling Recommendation** FOR ADDITIONAL INFORMATION
- SEE ILLUSTRATION 2-17 FOR MOUNTING INFORMATION

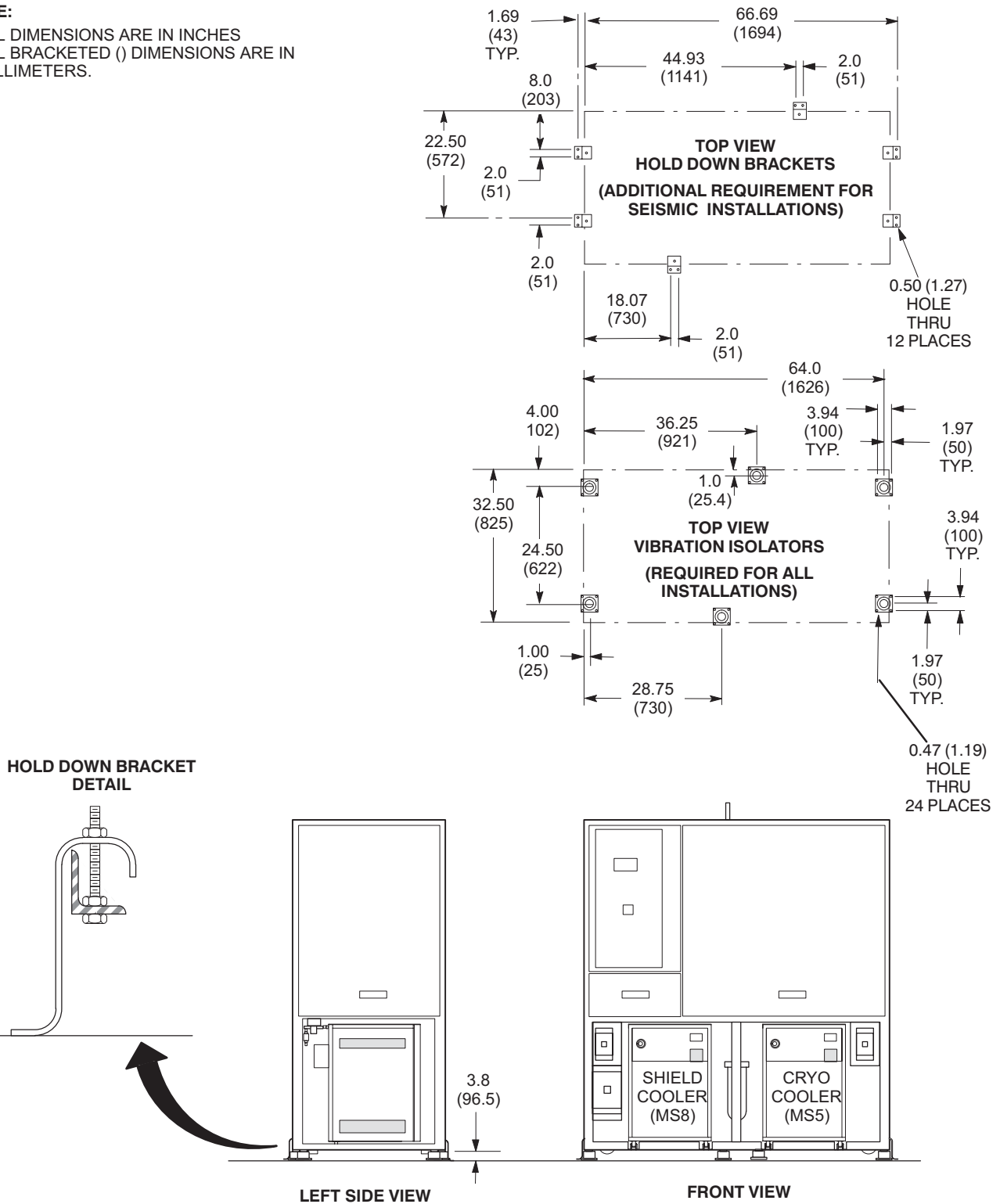


AIR COOLED SYSTEM COOLING CABINET [MODEL LC20M1] (SCC)

ILLUSTRATION 2-16

NOTE:

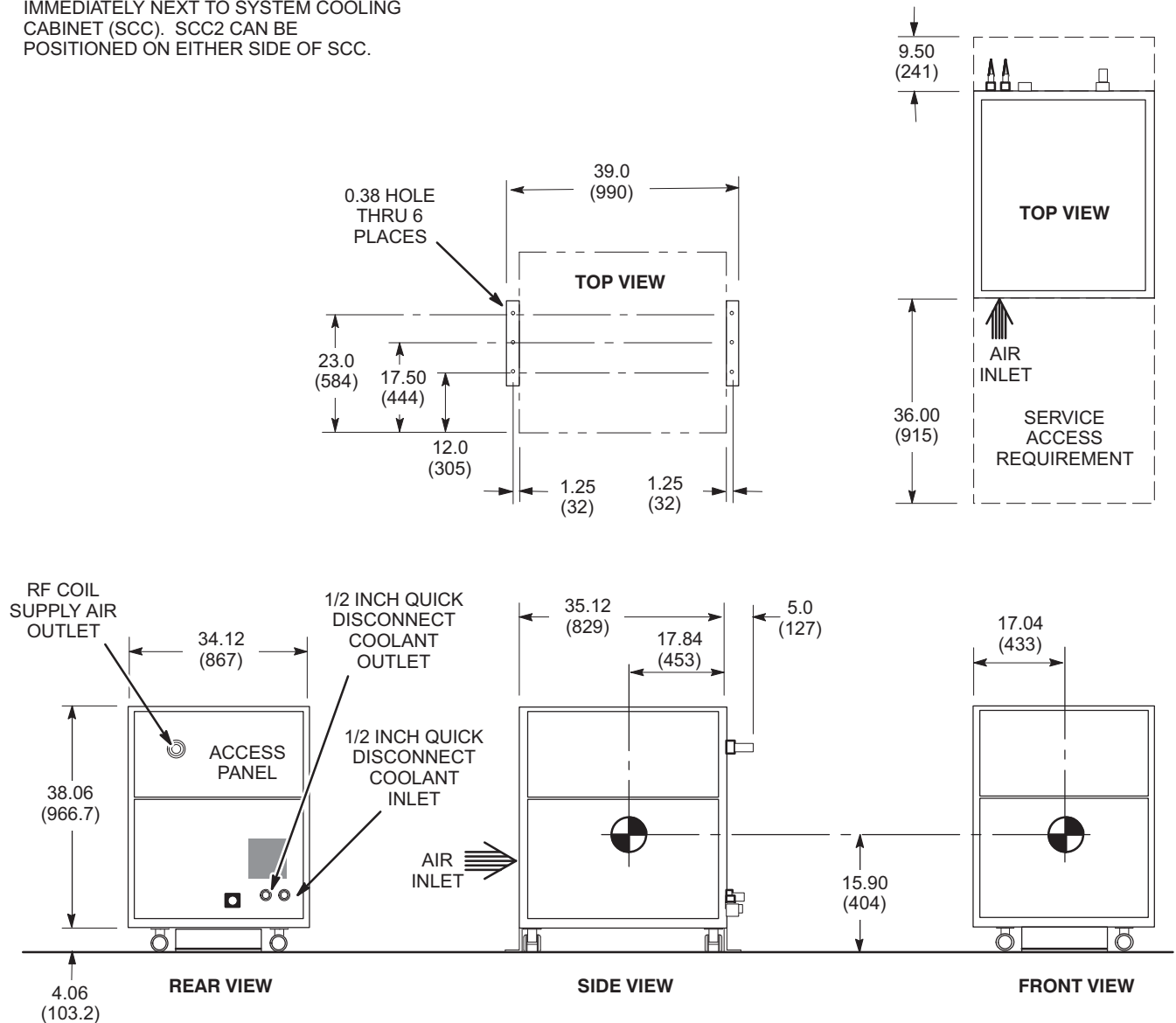
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



SYSTEM COOLING CABINET (SCC) MOUNTING
ILLUSTRATION 2-17

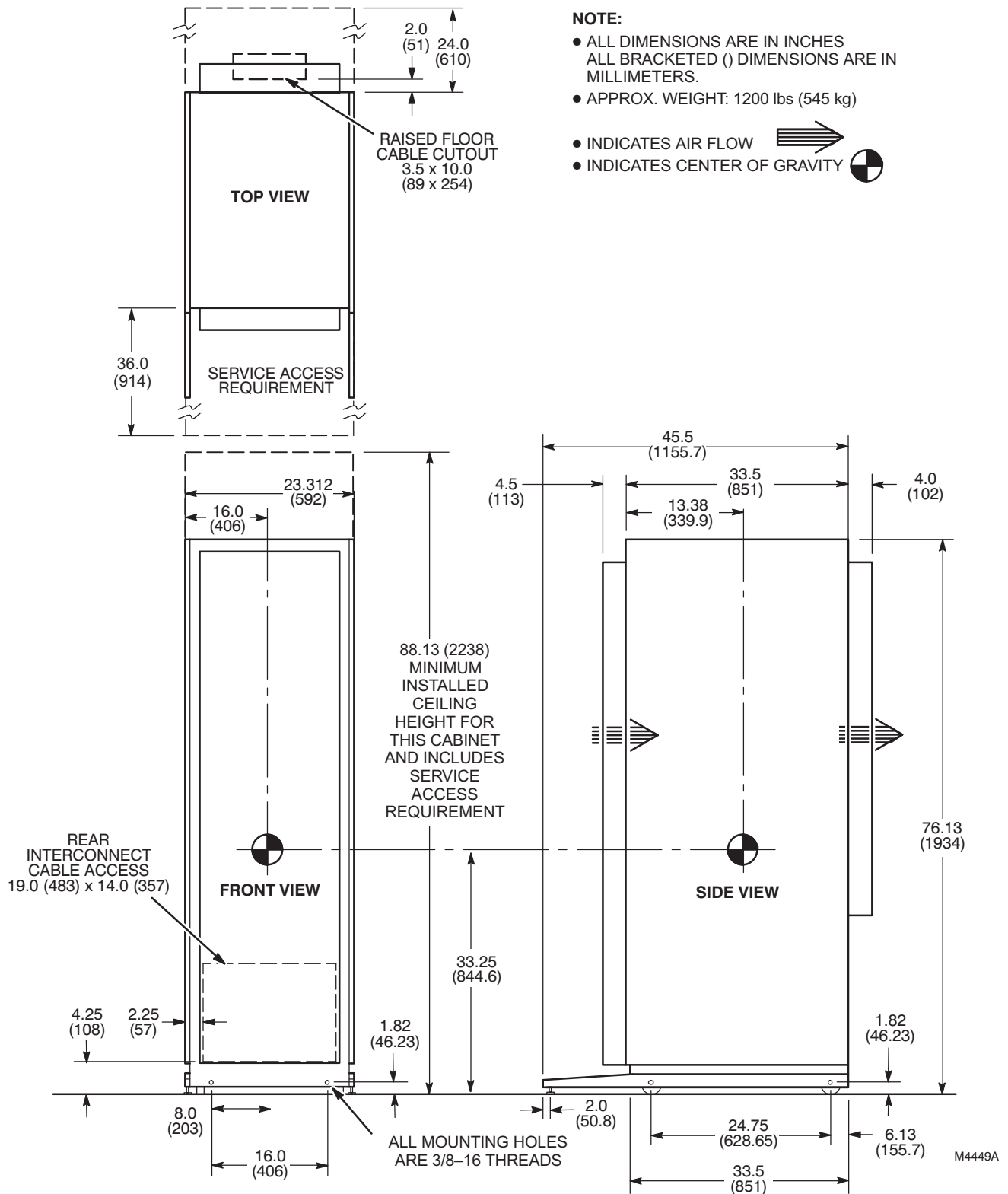
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 550 lbs (250 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR INLET 
- INDICATES CENTER OF GRAVITY 
- THIS CABINET MUST BE LOCATED IMMEDIATELY NEXT TO SYSTEM COOLING CABINET (SCC). SCC2 CAN BE POSITIONED ON EITHER SIDE OF SCC.



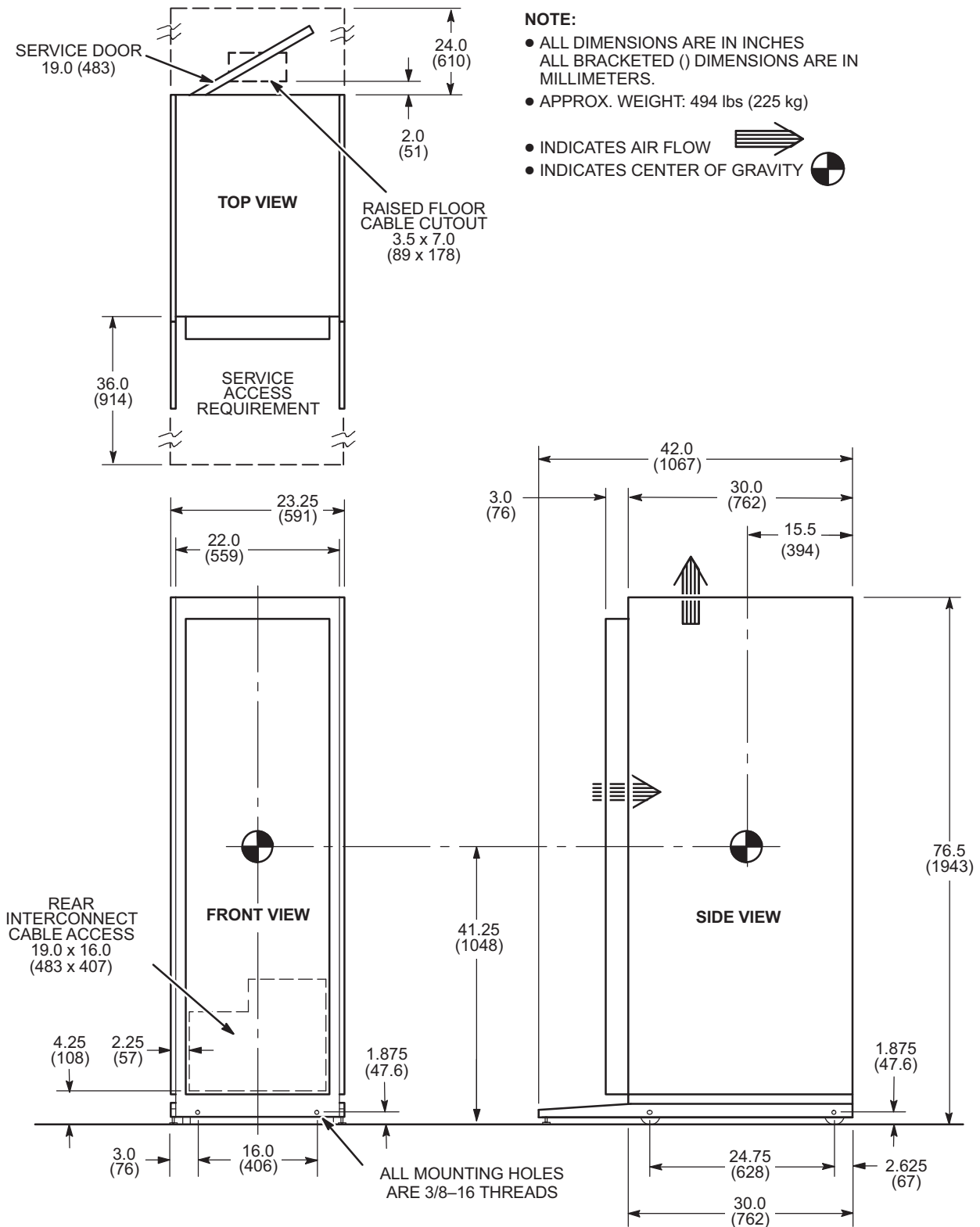
SYSTEM COOLING AUXILIARY CABINET [MODEL HPB-1] (SCC2)

ILLUSTRATION 2-18



POWER CABINET (MR1)

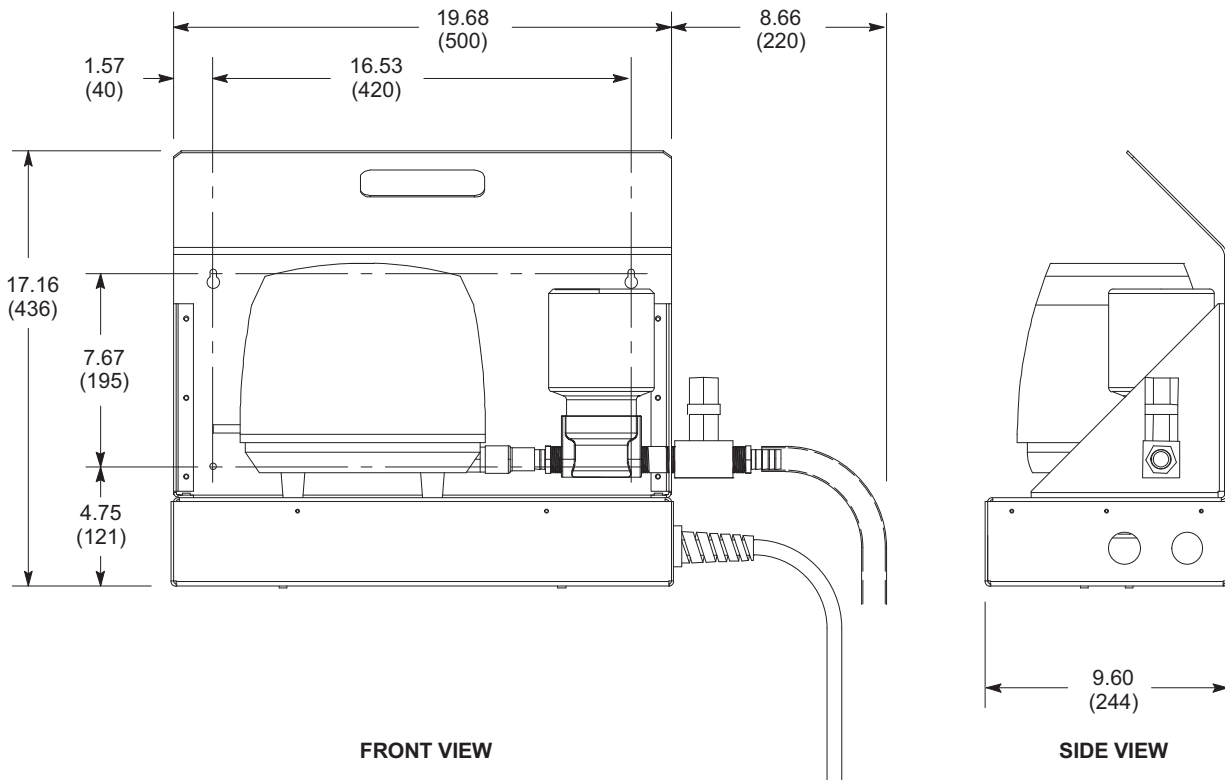
ILLUSTRATION 2-19



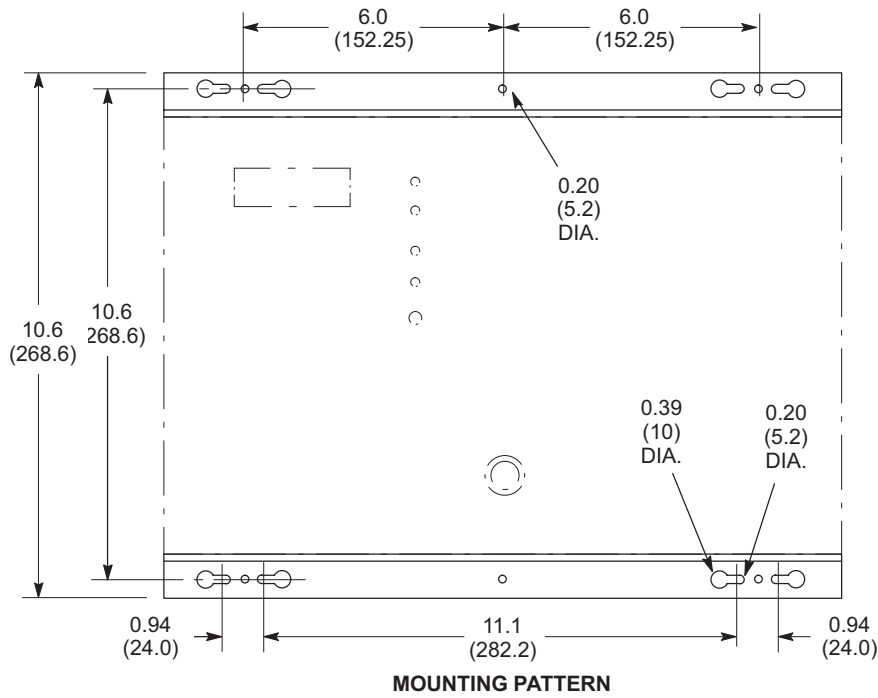
SYSTEM CABINET (MR2)
ILLUSTRATION 2-20

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (20.5 kg)

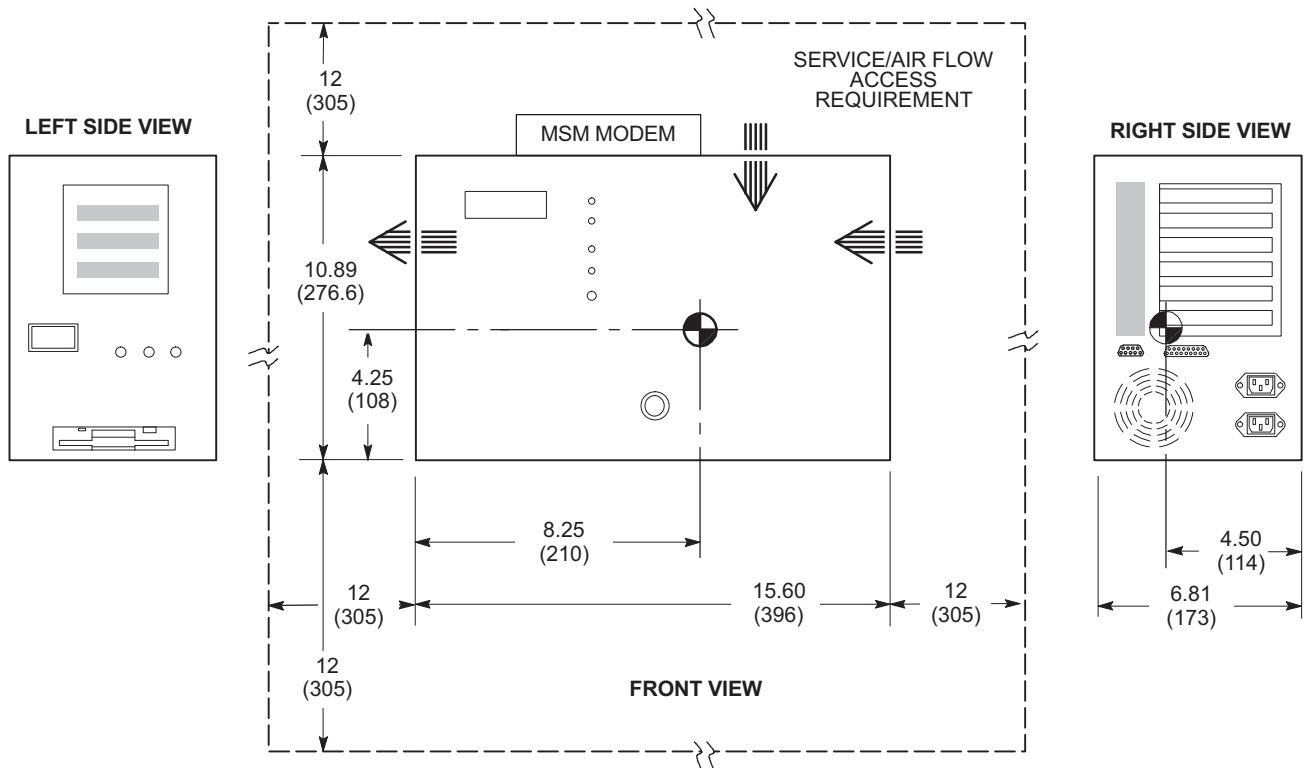


PATIENT COOLING COMPRESSOR
ILLUSTRATION 2-21



NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 22 lbs (10 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY



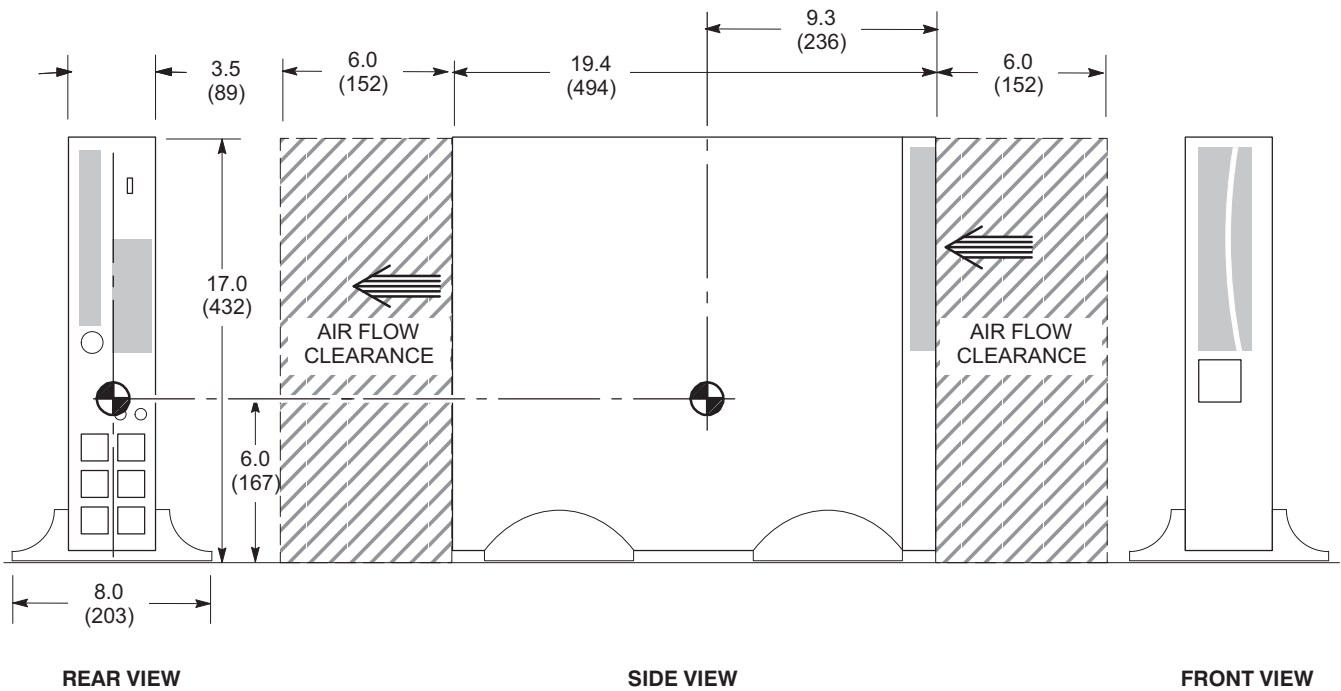
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.

- APPROX. WEIGHT: 50 lbs (23 kg)

- INDICATES AIR FLOW

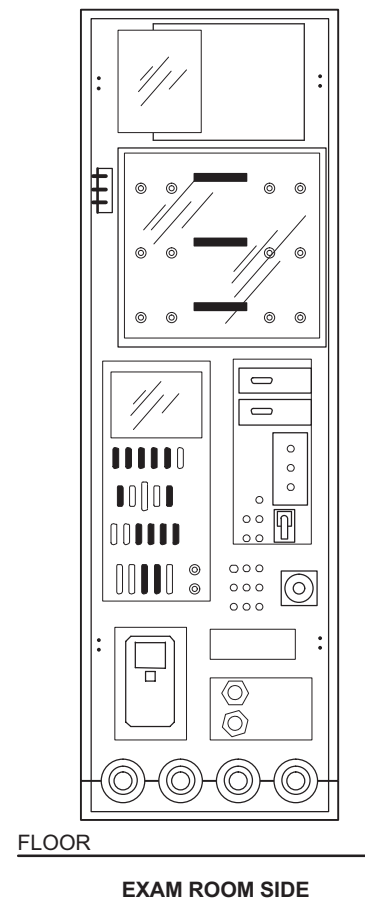
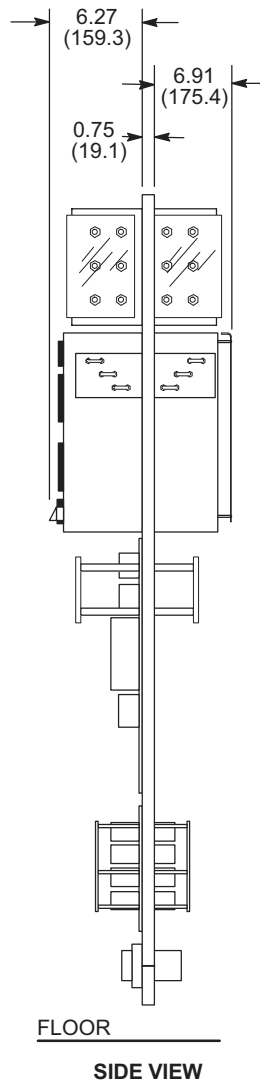
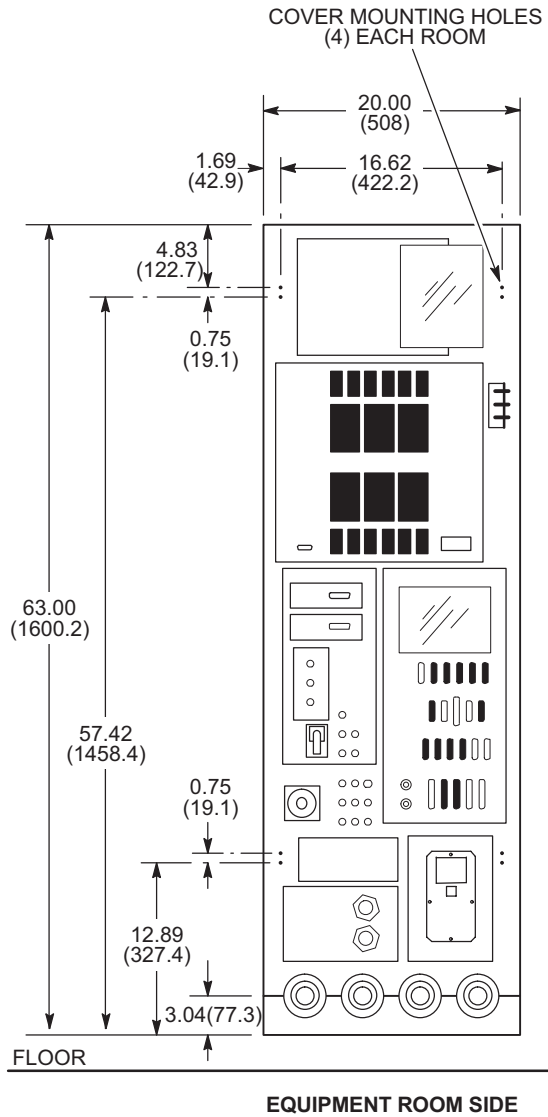
- INDICATES CENTER OF GRAVITY



UPS FOR MAGNET MONITOR
ILLUSTRATION 2-23

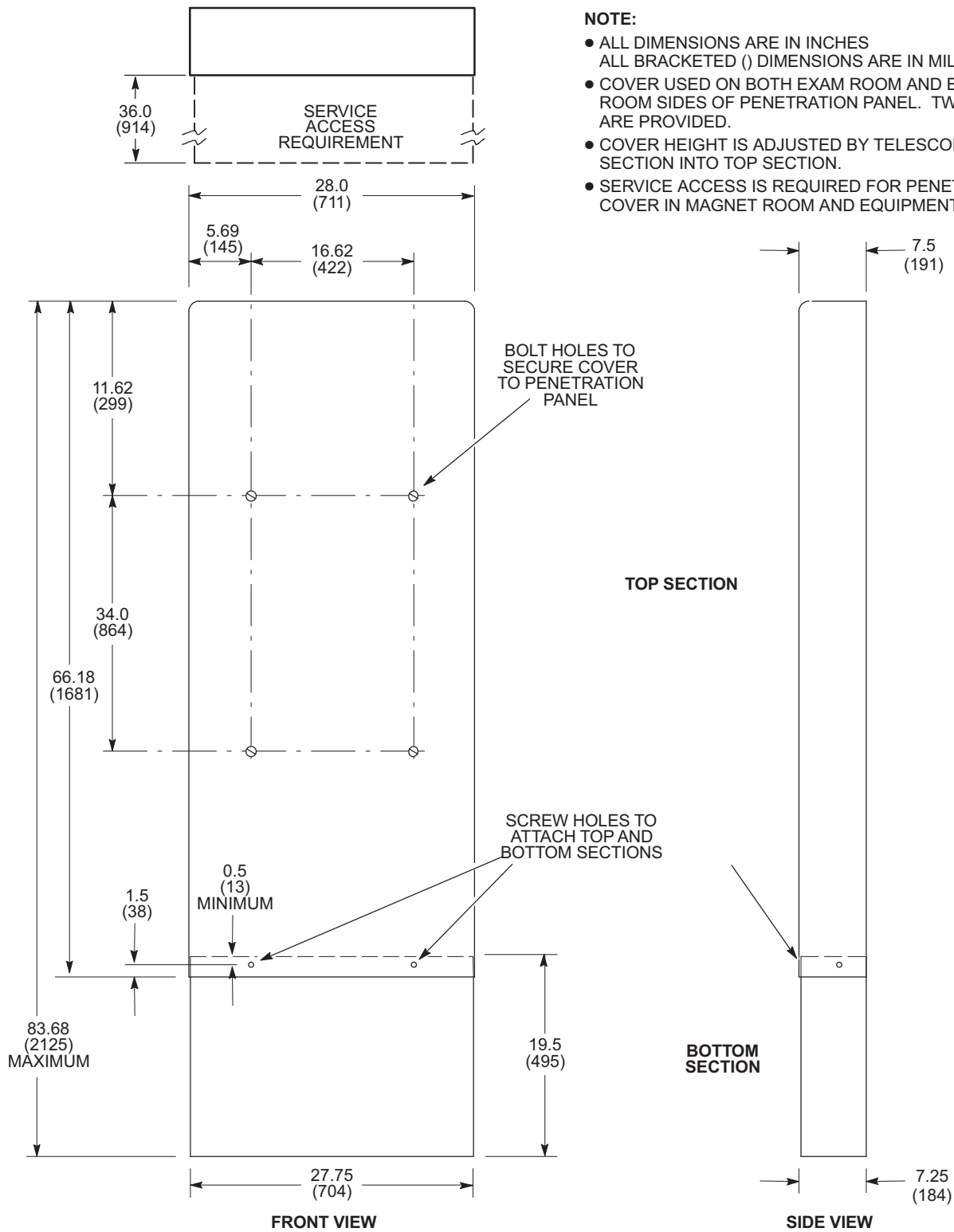
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROXIMATE WEIGHT: 52 lbs (23.6 kg)
- SEE ILLUSTRATION 2-25 FOR SERVICE ACCESS
REQUIRED FOR PENETRATION PANEL & COVERS IN
MAGNET ROOM AND EQUIPMENT ROOM.



M4528A

PENETRATION PANEL (PP1)
ILLUSTRATION 2-24

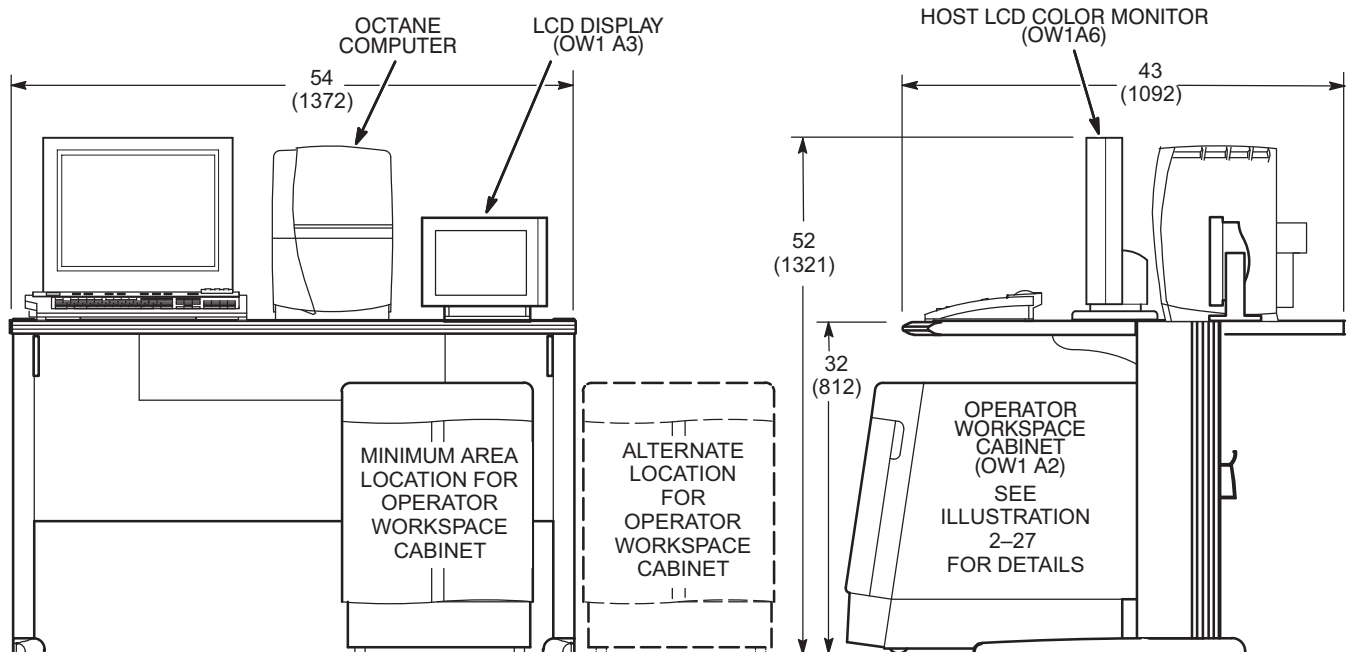
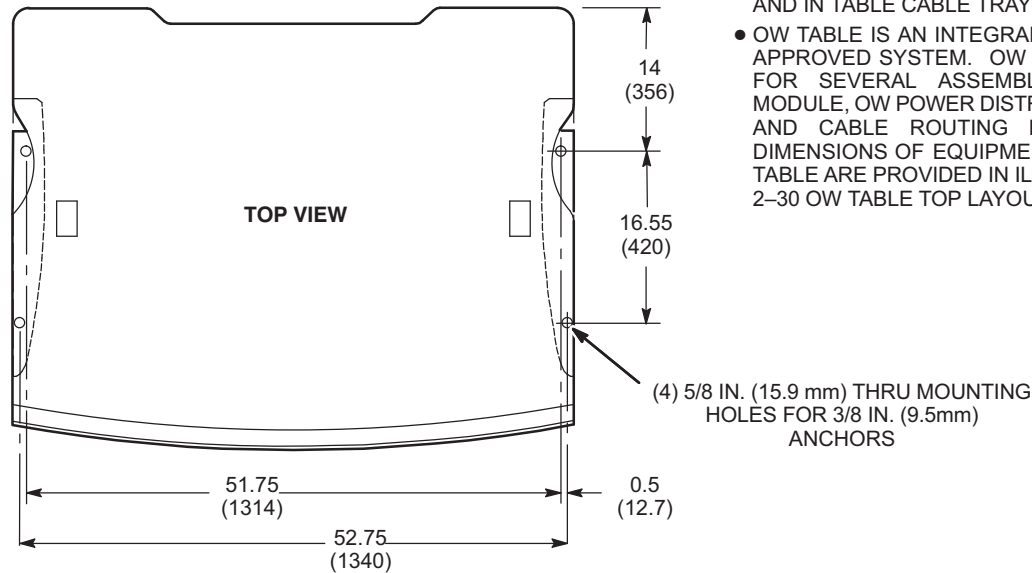


M4009A1M

PENETRATION PANEL COVER
ILLUSTRATION 2-25

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- ASSEMBLIES WHICH MOUNT TO UNDERSIDE OF TABLE AND IN TABLE CABLE TRAY ARE NOT SHOWN.
- OW TABLE IS AN INTEGRAL PART OF THE REGULATORY APPROVED SYSTEM. OW TABLE PROVIDES MOUNTING FOR SEVERAL ASSEMBLIES (E.G. OW INTERFACE MODULE, OW POWER DISTRIBUTION BOX, MODEM, DASM) AND CABLE ROUTING FOR OW INTERCONNECTS. DIMENSIONS OF EQUIPMENT LOCATED ON TOP OF OW TABLE ARE PROVIDED IN ILLUSTRATIONS 2-28 THROUGH 2-30 OW TABLE TOP LAYOUT PLANNING PURPOSES.



FRONT VIEW

SIDE VIEW

OPERATOR WORKSPACE (OW1)

ILLUSTRATION 2-26

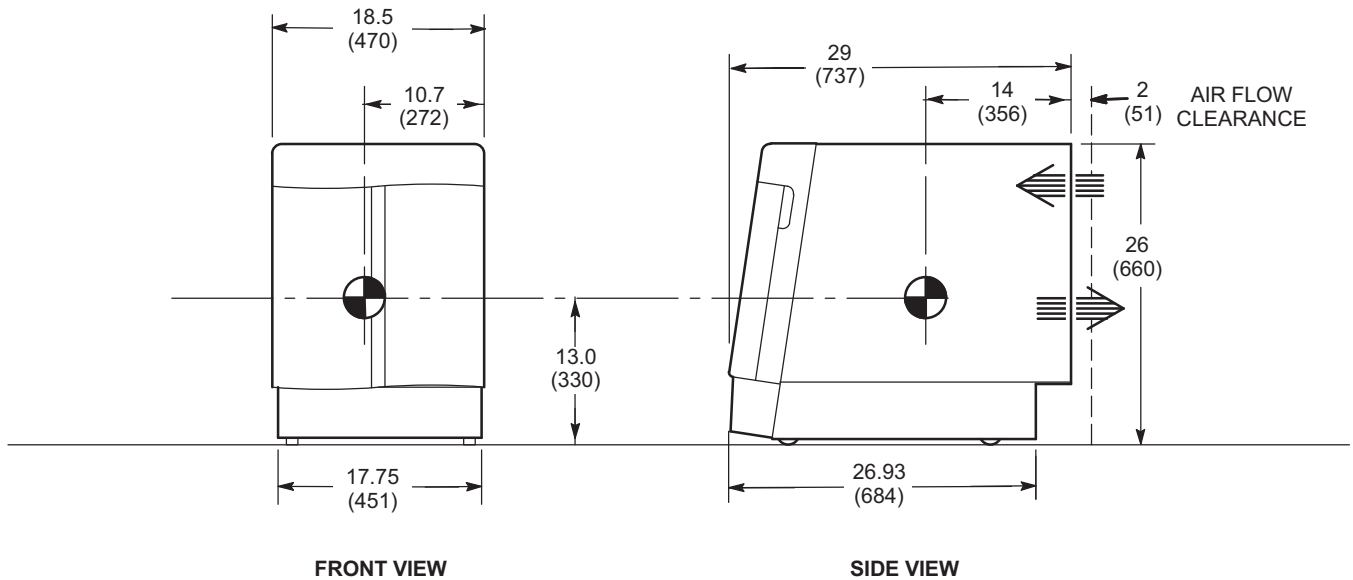
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 192 lbs (87 kg)

- INDICATES AIR FLOW

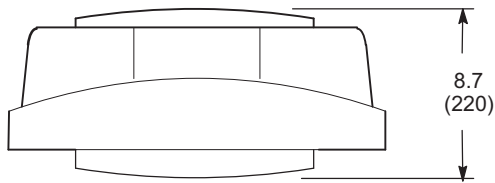


- INDICATES CENTER OF GRAVITY

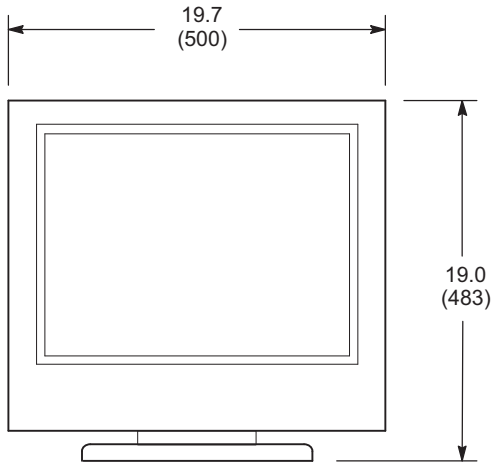


OPERATOR WORKSPACE CABINET (OW1 A2)

ILLUSTRATION 2-27



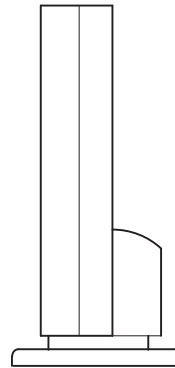
TOP VIEW



FRONT VIEW

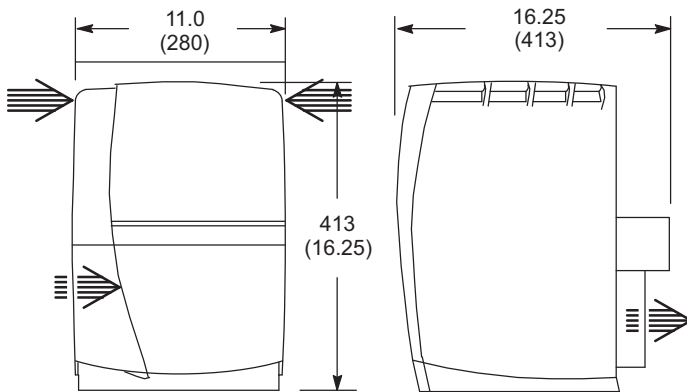
NOTE:

- ALL DIMENSIONS ARE IN INCHES
- ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



SIDE VIEW

OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR
ILLUSTRATION 2-28



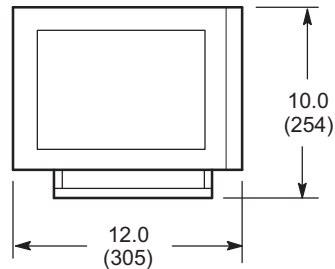
FRONT VIEW

SIDE VIEW

OCTANE COMPUTER

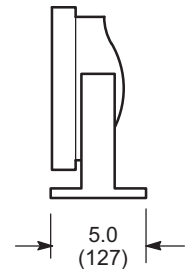
NOTE:

- ALL DIMENSIONS ARE IN INCHES
- ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- INDICATES AIR FLOW ➡



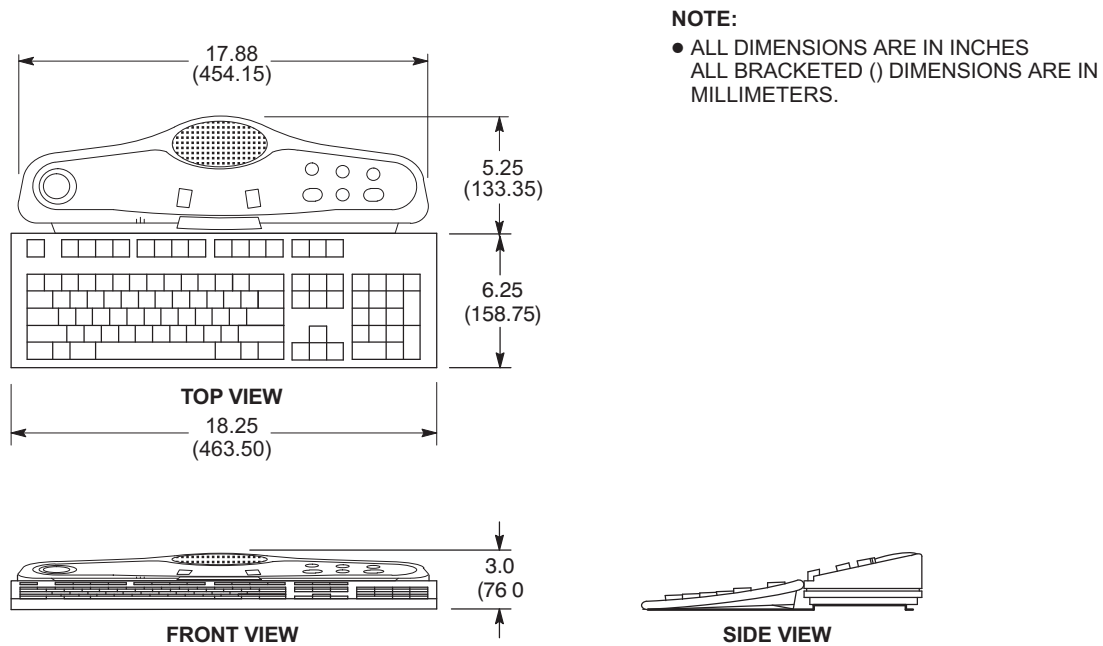
FRONT VIEW

LCD DISPLAY

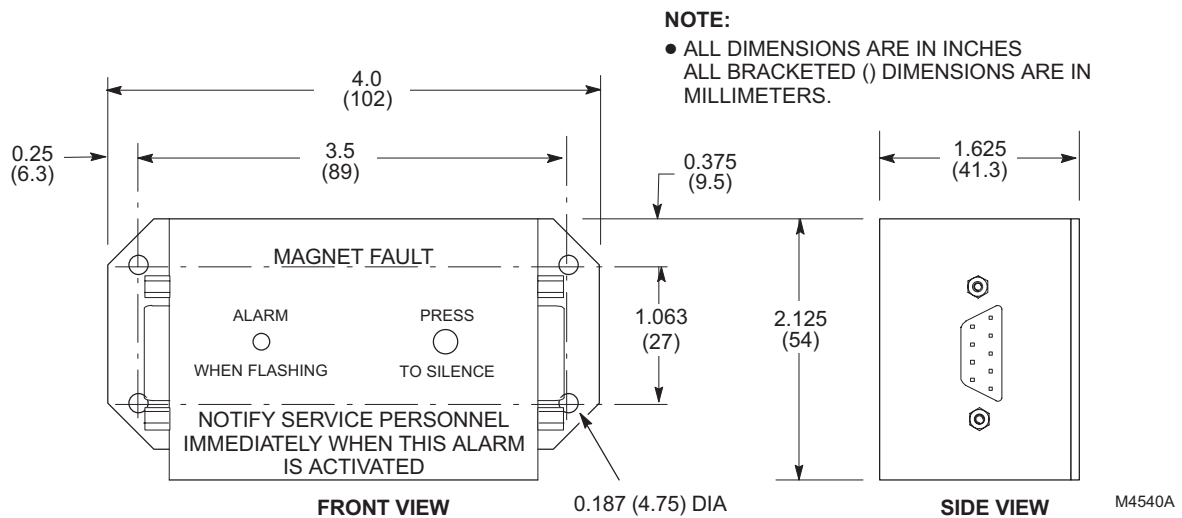


SIDE VIEW

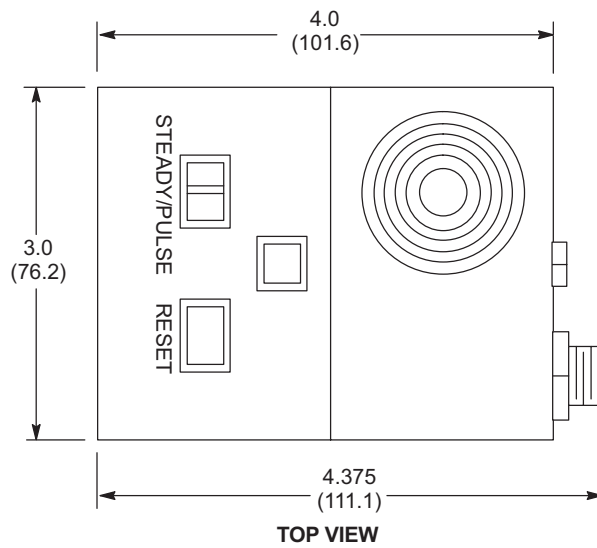
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER
ILLUSTRATION 2-29



OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – KEYBOARD
ILLUSTRATION 2-30

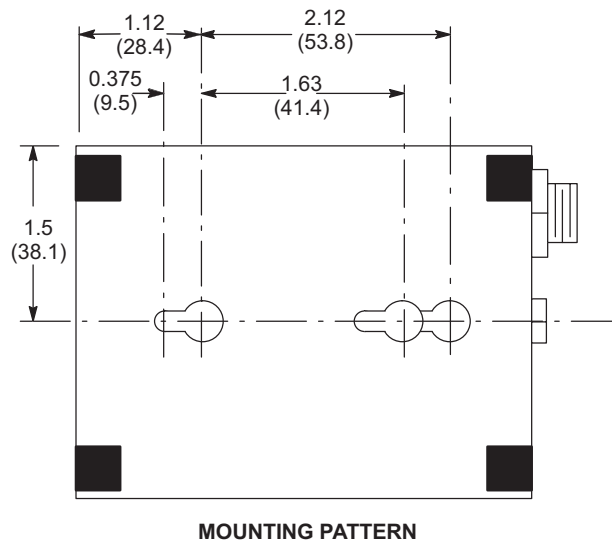
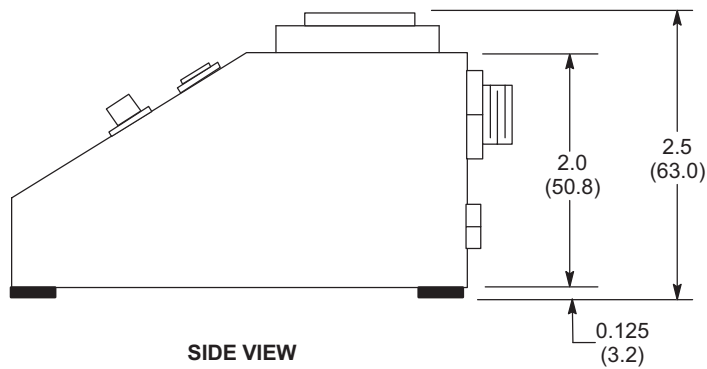


MAGNET MONITOR REMOTE ALARM BOX (MSM2) POSITIONED ON OW TABLE TOP
ILLUSTRATION 2-31



NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 0.5 lbs (0.2 kg)



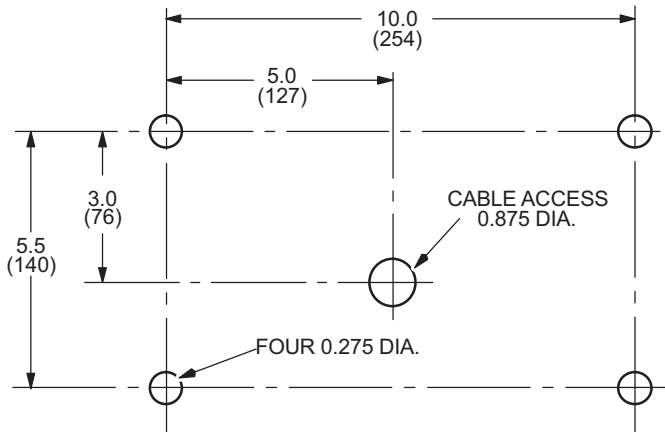
PNEUMATIC PATIENT ALERT CONTROL BOX (PA1)

ILLUSTRATION 2-32

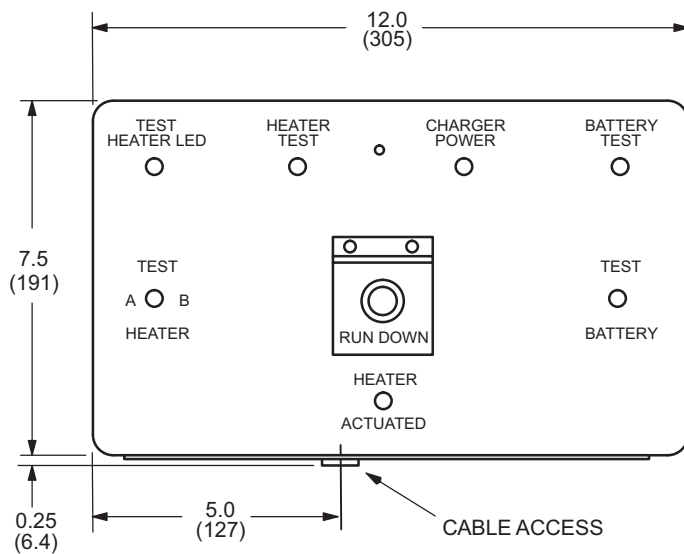
M4263A1

NOTE:

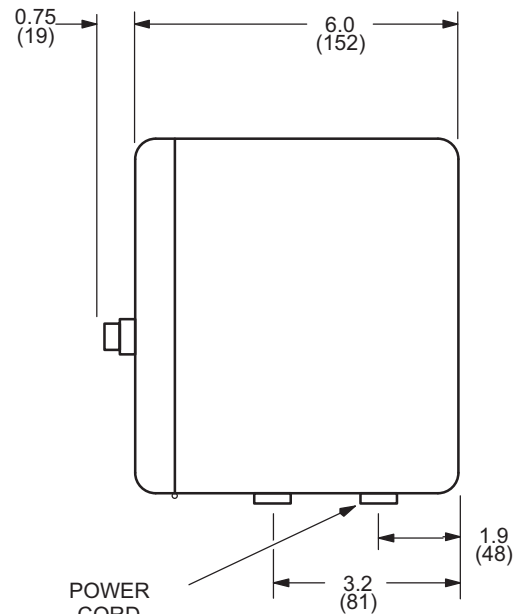
- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 8.8 lbs. (4kg)



MOUNTING PATTERN



FRONT VIEW



SIDE VIEW

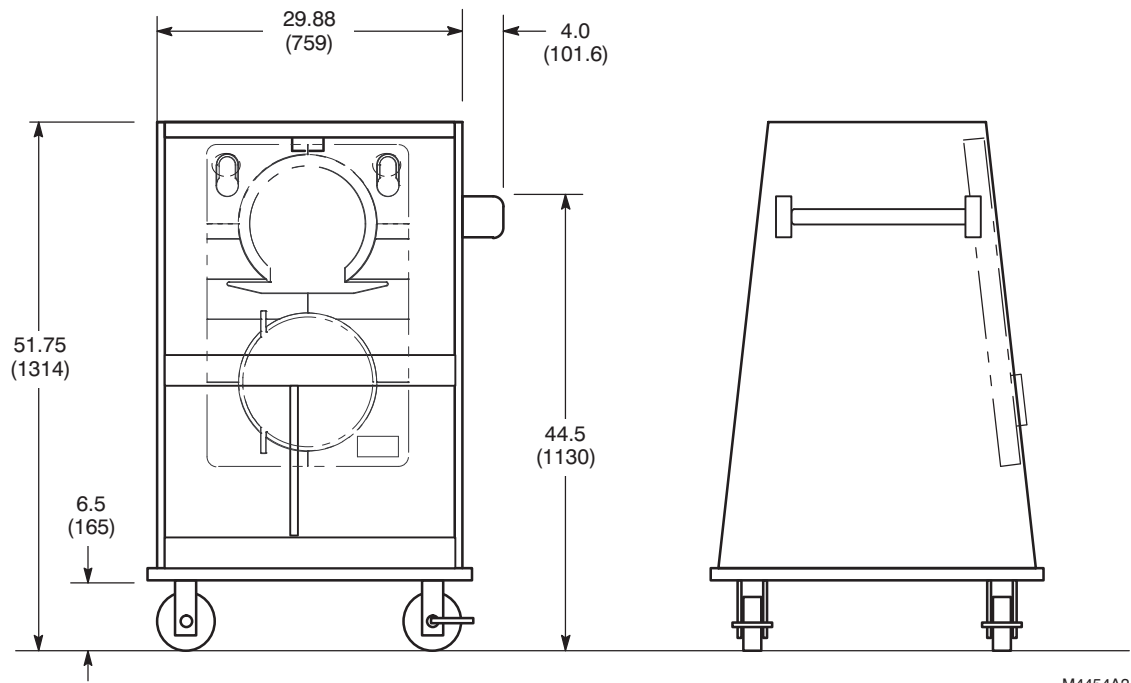
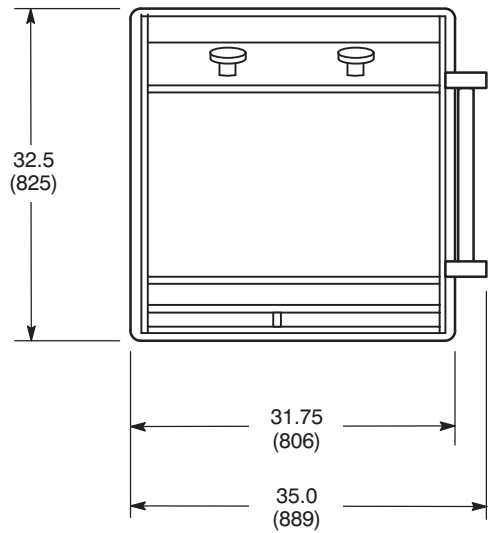
M1517A5

MAGNET RUNDOWN UNIT (MS4)

ILLUSTRATION 2-33

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE
IN MILLIMETERS.
- APPROX. WEIGHT INCLUDING
PHANTOMS: 350 lbs (159 kg)

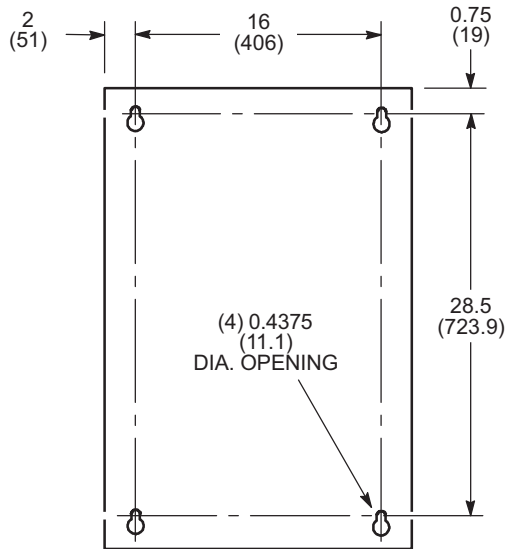


M4454A2

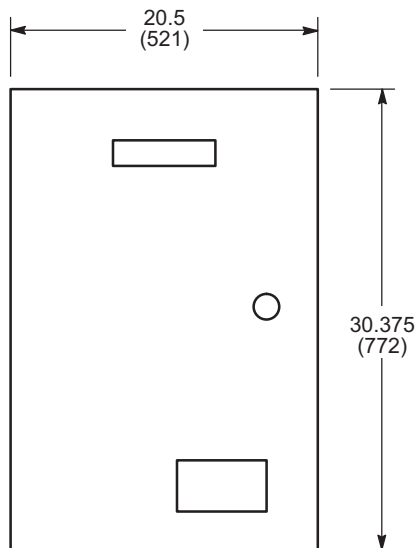
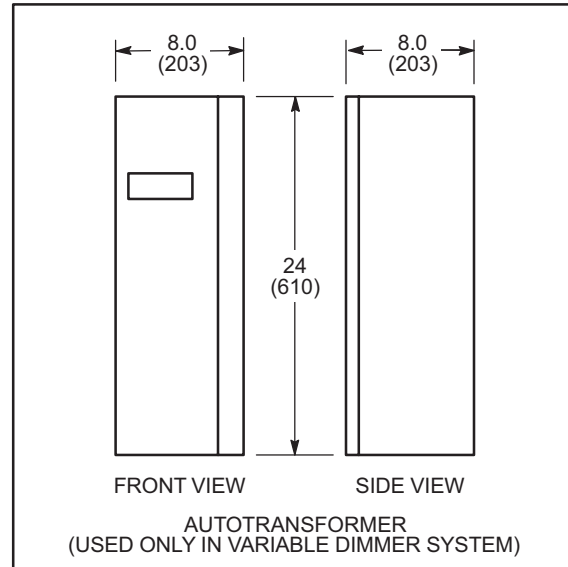
SPT PHANTOM SET SHIPPING/STORAGE CART
ILLUSTRATION 2-34

NOTE:

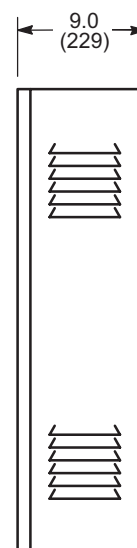
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT
CONTROL PANEL: 155 lbs (70 kg)
AUTOTRANSFORMER: 60 lbs (27 kg)
- DC POWERED LIGHTING IS REQUIRED IN THE MAGNET ROOM PER SECTION 4-6 LIGHTING. ALSO REFER TO SECTION 1-4 FACILITY OPTIONS AND SECTION 5-8 DC LIGHTING CONTROLLER.



MOUNTING PATTERN
(CONTROL PANEL)



FRONT VIEW
(CONTROL PANEL)



SIDE VIEW
(CONTROL PANEL)

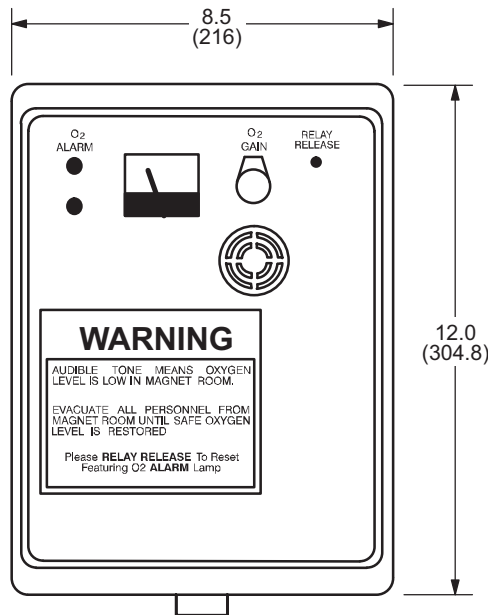
M1519A3M

DC LIGHTING CONTROLLER – GE FACILITY OPTION

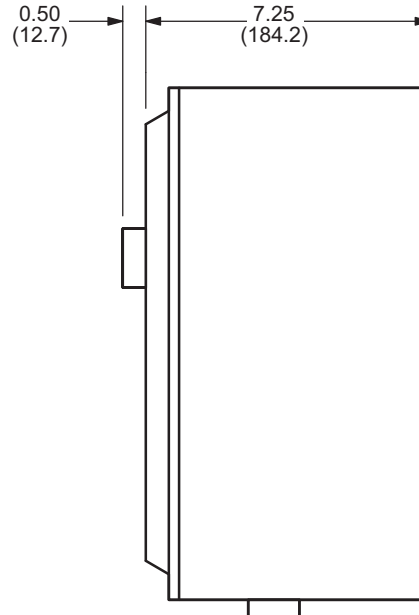
ILLUSTRATION 2-35

NOTE:

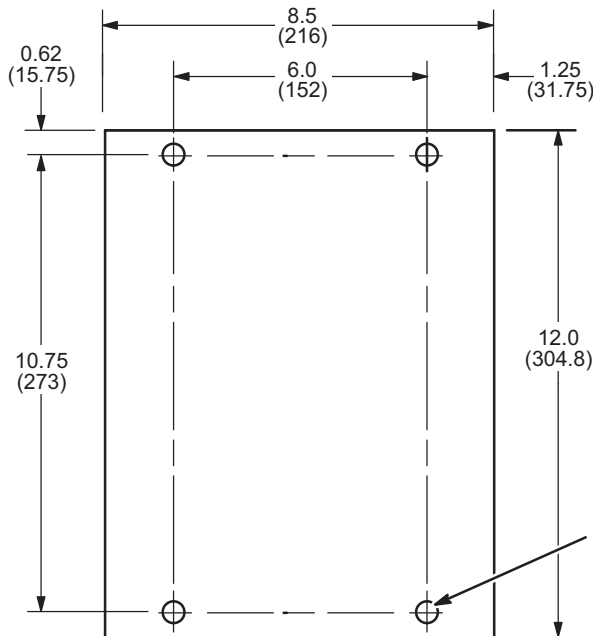
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 9 lbs (4.1 kg)



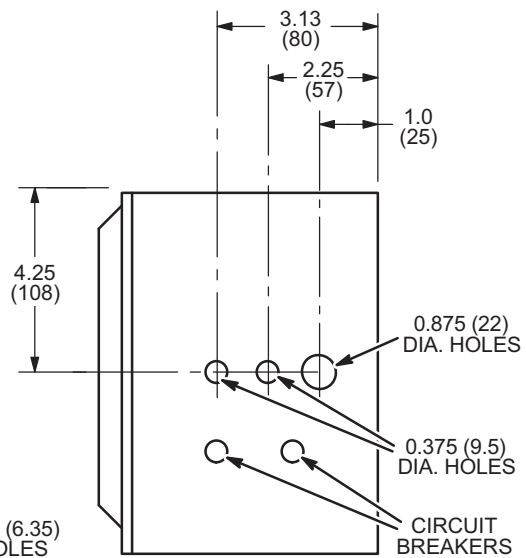
FRONT VIEW



SIDE VIEW



MOUNTING PATTERN



BOTTOM VIEW

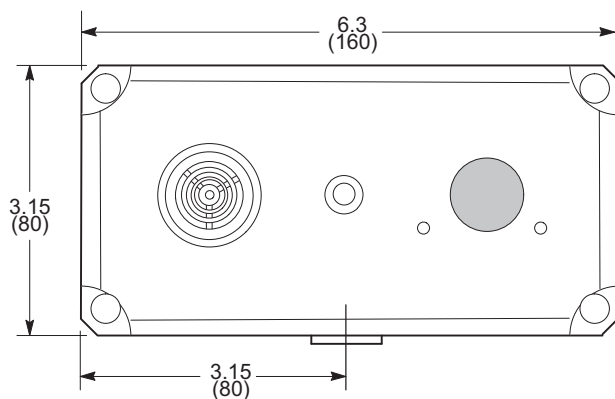
**OXYGEN MONITOR (OM1) – OPTIONAL
ILLUSTRATION 2-36**

M1516A4

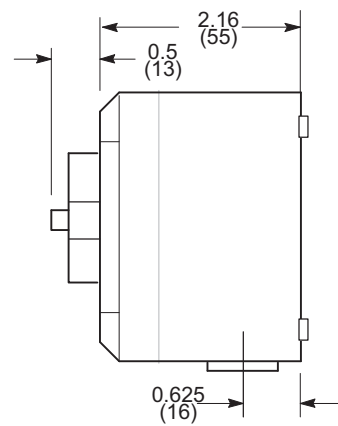


NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1.5 lbs (0.68 kg)



FRONT VIEW



SIDE VIEW

REMOTE OXYGEN SENSOR MODULE (OM3) – OPTIONAL

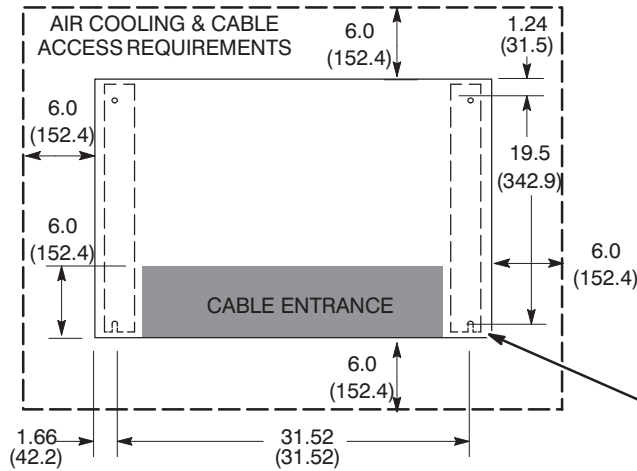
ILLUSTRATION 2-37

M3346A1

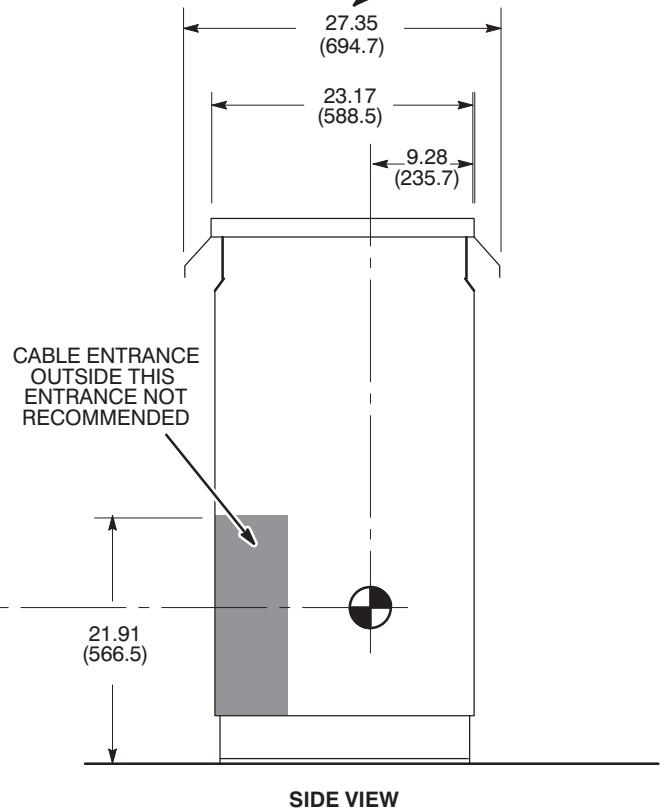
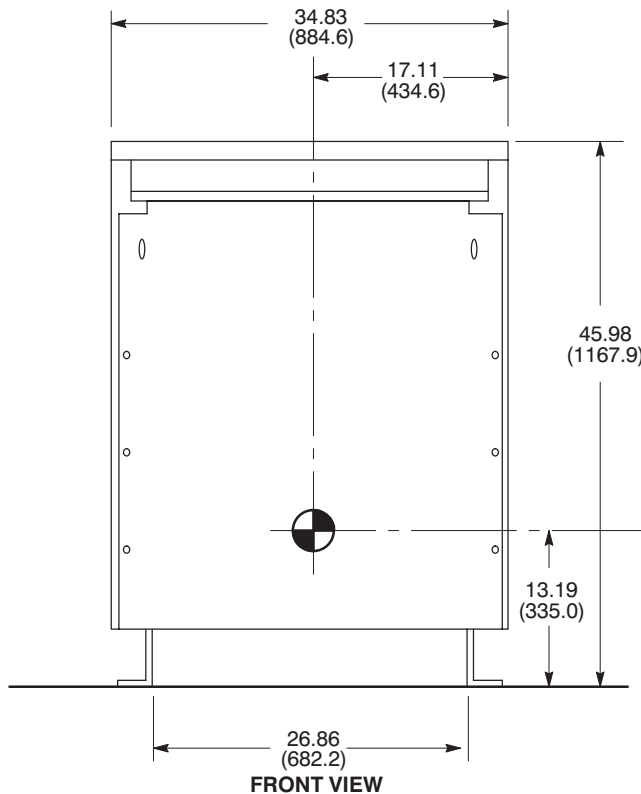
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 875 lbs (397 kg)

- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 



TOP VIEW AND MOUNTING FOOTPRINT

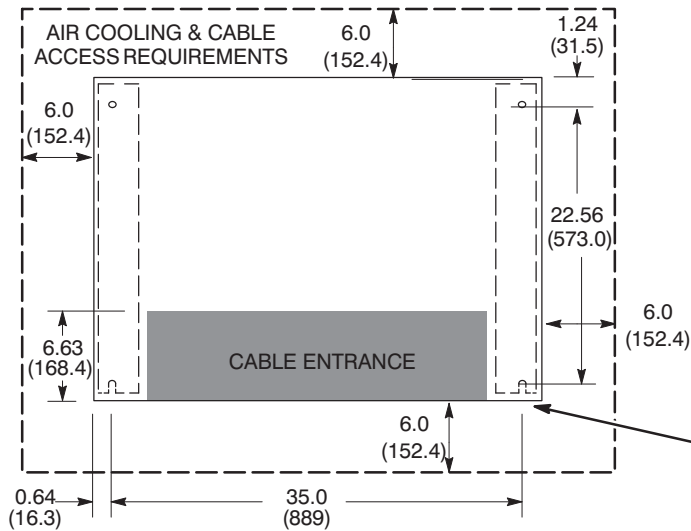


150 KVA 208-480Y277 VOLT, 60 Hz GE TRANSFORMER – R4500AS

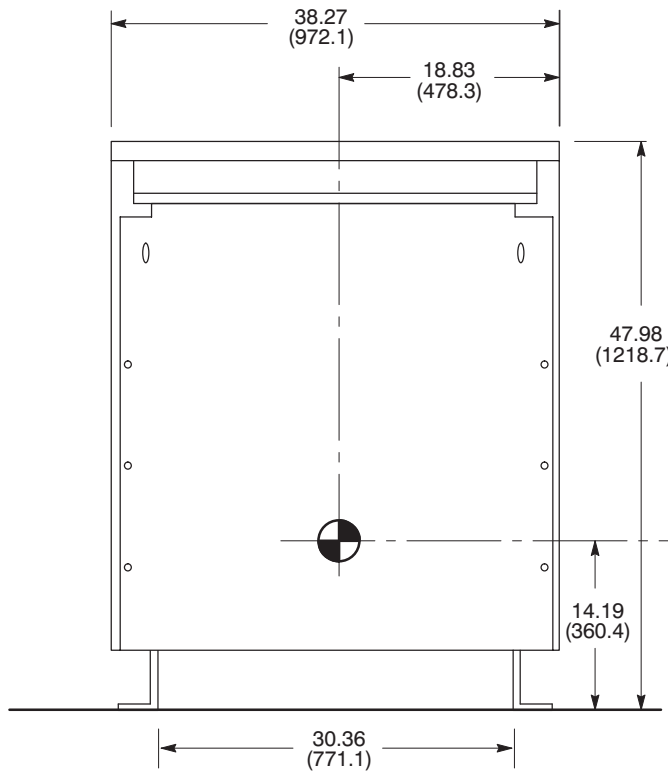
ILLUSTRATION 2-38

NOTE:

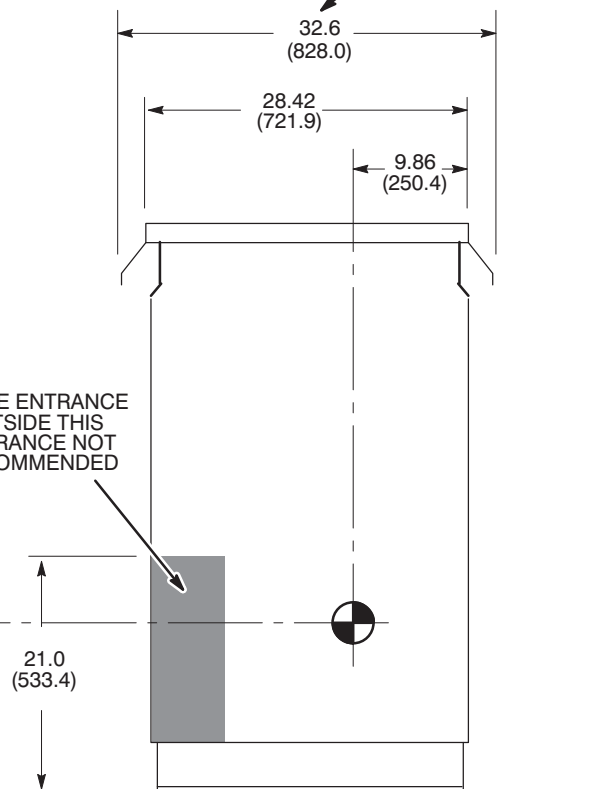
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1160 lbs (526.3 kg)
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 



TOP VIEW AND MOUNTING FOOTPRINT



FRONT VIEW



SIDE VIEW

150 KVA 200–400Y230 VOLT, 50/60 Hz GE TRANSFORMER – R4500BD

ILLUSTRATION 2–39